



4/3- and 4/2- way Directional Control Valves Model: 4DE10 with Wet Pin AC or DC Solenoids

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Y02101/07.06

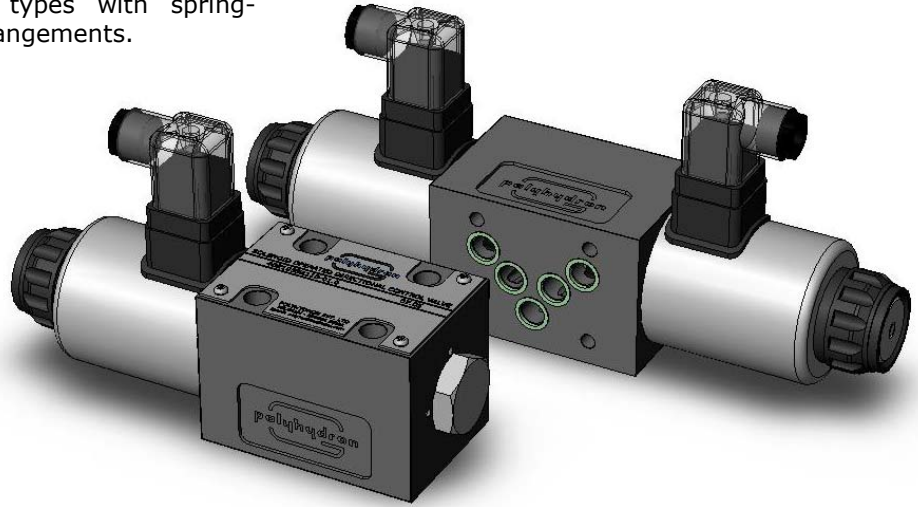
Size 10 (D 05)

315 bar

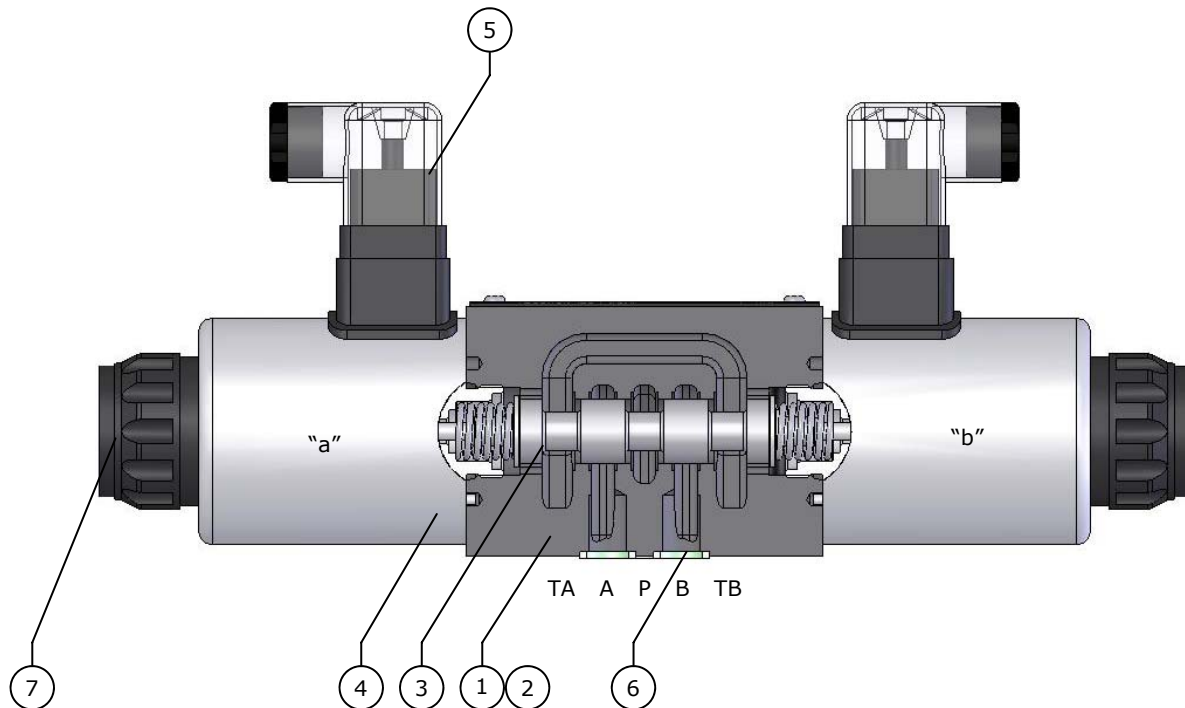
120lpm

Description

4-port, spool type, solenoid operated, directional control valves with wet armature AC or DC coils and wide variety of spool types with spring-centered and spring-offset arrangements.



Features



1. Five chamber body and spool design to facilitate high flow.
2. Conforms to ISO4401-5, NFPA T3.5.1M R1 and ANSI B93-7 D 05 interface.
3. Available in 52 standard, interchangeable spool configurations.
4. Removable wet-armature AC and DC solenoids for quick replacement and conversions. Moulded solenoid coils to protect against moisture and ingress of dust.

5. Individual solenoid plug-in connectors with indicator lights to ISO 4400 and DIN 43650.
6. Special R-rings for high extrusion resistance. Increased stability under high dynamic condition.
7. Override for operation of valve without electrical power.

Note on solenoid identification:

Solenoid "a" is close to Port "A" and Solenoid "b" is close to Port "B" for a direct operated valve.



4/3- and 4/2- way Directional Control Valves
Model: 4DE10 with Wet Pin AC or DC Solenoids

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Size 10 (D 05)

315 bar

120lpm

Y02101/07.06

Technical data

General

Weight		AC	DC
Valve with 1 solenoid	kg	4.2	4.3
Valve with 2 solenoids	kg	5.7	5.8

Hydraulic

Nominal Pressure		
Ports P, A and B	bar	315
Port T	bar	175

Maximum Flow	lpm	120
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Flow Area

Spool Q, A/B → T, position "0"	6%	5.9mm ²
Spool R, B → P, position "b"	50%	49.3mm ²
Spool V, A/B → T, position "0"	12%	11.8mm ²
P → A/B, position "0"	10%	9.9mm ²
Spool W, A/B → T, position "0"	3%	3.0mm ²
Others		>98.0mm ²

Hydraulic Fluid	Mineral Oil
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Fluid Temperature		
NBR	°C	-30 to 80°C
FKM	°C	-20 to 80°C

Maximum degree of fluid contamination	Class 18/15 according to ISO 4406
	Filter $\beta_{10} \geq 75$

Electrical

Power Requirement	W	35
Duty Cycle		Continuous
Shifting Time		
ON	ms	60
OFF	ms	30
Insulation		IP 65

Spool Types

4/2 Way Valve with "0" and "a" positions



A⁽³⁾



C



D



4/2 Way Valve with "0" and "b" positions



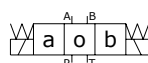
B⁽³⁾



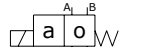
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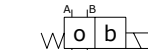
4/3 Way and 4/2⁽¹⁾ Way Valve



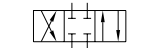
*A



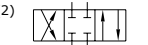
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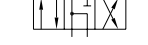
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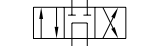
E1⁽²⁾



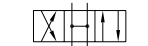
F



G



H



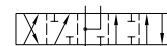
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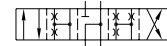
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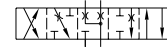
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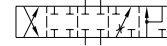
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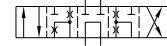
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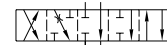
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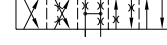
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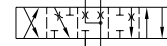
U



V



W



Note:

⁽¹⁾ When ordering a spool with only two positions "o" and "a" or "o" and "b", specify the desired position "A" or "B" after the spool code.
Example: E Spool with "o" and "a" position—4DE10EA...

⁽²⁾ Please check the possibility of cylinder intensification

⁽³⁾ For spool type A and B. Port T must be used as drain line



4/3- and 4/2- way Directional Control Valves
Model: 4DE10 with Wet Pin AC or DC Solenoids

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Size 10 (D 05)

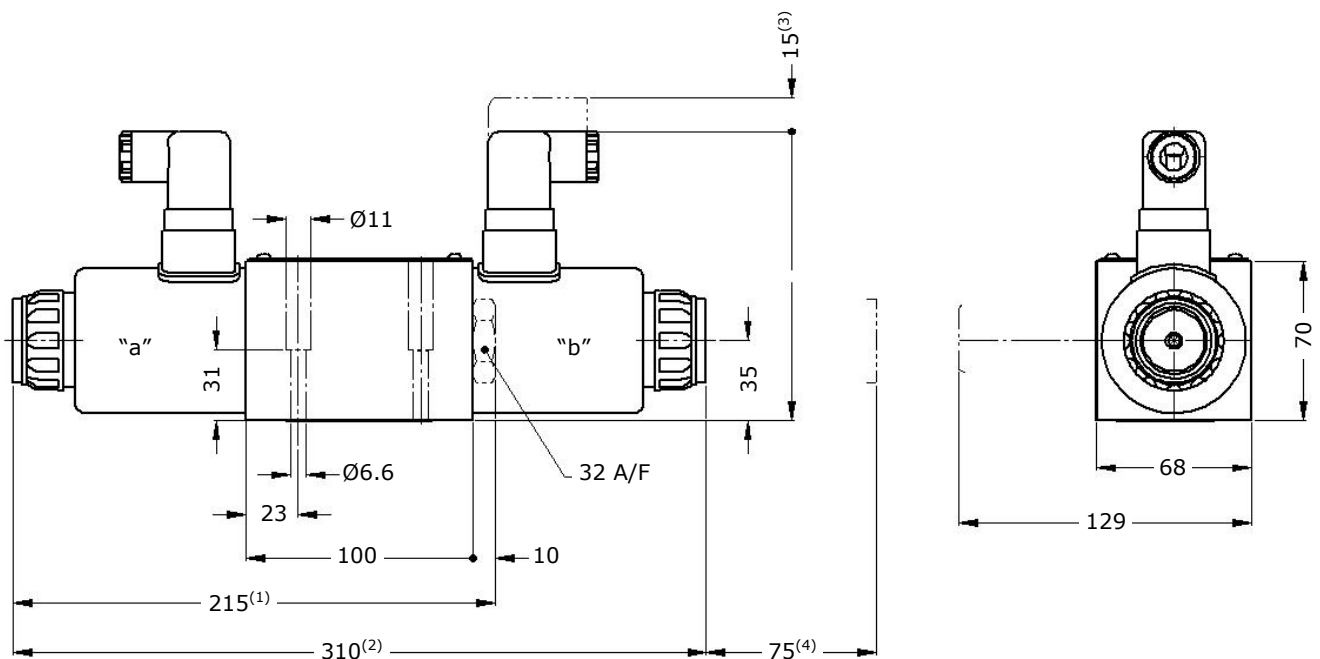
315 bar

120lpm

Ordering Code

4	DE	10	E	S	A220	V	-01	
4 Service Ports				Design Series (Subject to Change) 01 to 09 externally interchangeable				
Directional Control, Electrically Operated				Seals				
ISO size 5, NFPA/ANSI D 05 interface				Nitrile (NBR) = omit				
Spool type (Refer Spool Chart)				Fluorocarbon (FKM) = V				
Return				Voltage				
Spring return = S				AC Voltage		DC Voltage		
Without spring return = O				A110		D12		
				A220		D24		

Installation

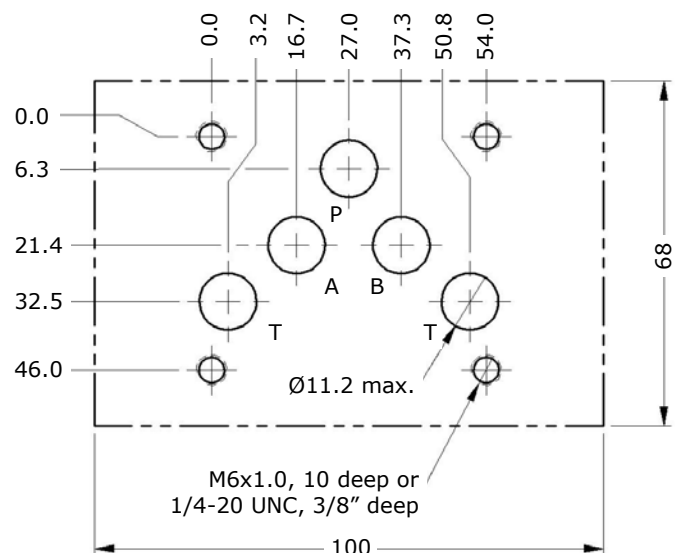
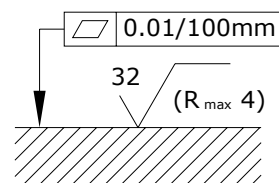


Note:

- (1) Length of valve with 1 solenoid
- (2) Length of valve with 2 solenoids
- (3) Space required to remove plug-in connector
- (4) Space required to remove solenoid

Valve Mounting Bolts:

M6 x 40 mm or 1/4-20 UNC x 1.1/2"
Tightening Torque 15.5 Nm
Mounting Hardware not included in scope of supply





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Size 10 (D 05)

315 bar

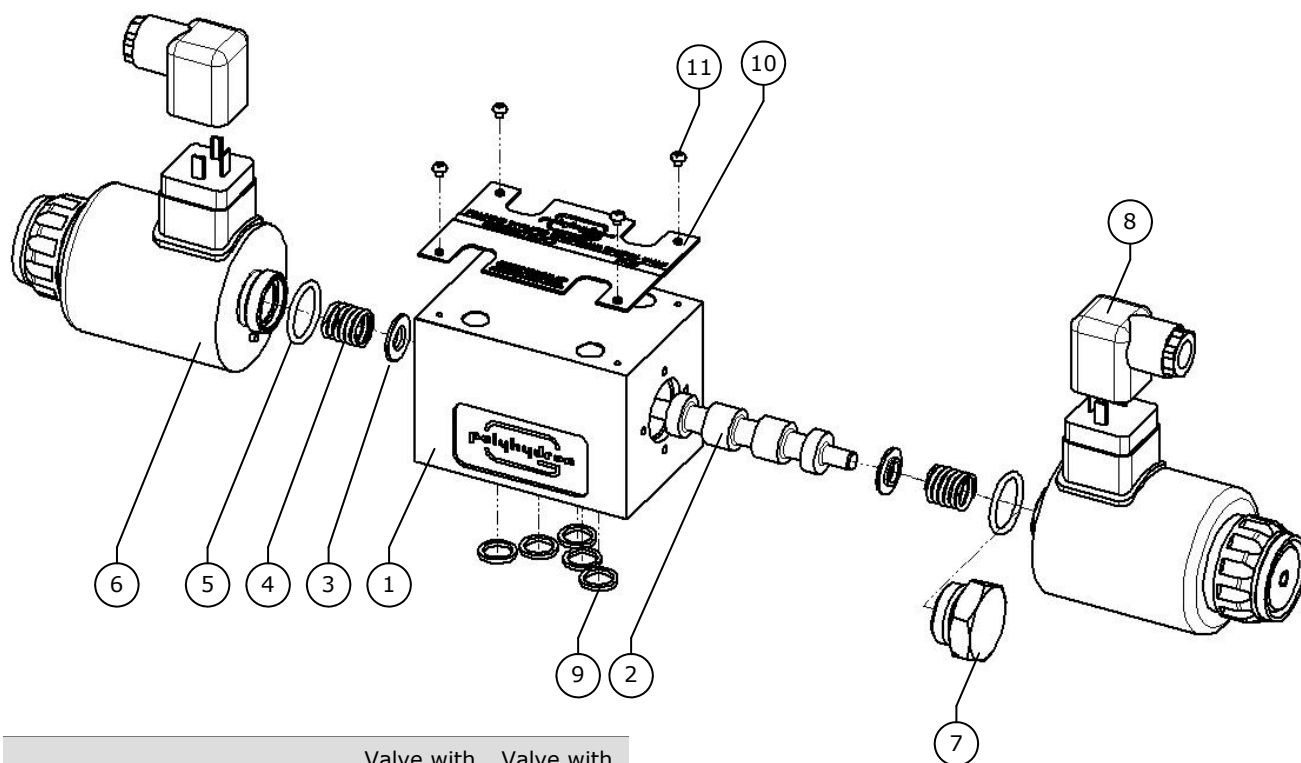
120lpm

ENGINEERING

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Y02101/07.06

Spares



Item	Description	Valve with 1 solenoid	Valve with 2 solenoids
1	Valve Body	1	1
2	Spool	1	1
3	Spring Pad	2	2
4	Spring	2	2
*	5 O-ring 21.9 x 2.62	2	2
6	Solenoid Coil	1	2
7	Plug	1	0
*	8 Plug-in connector	1	2
*	9 R-ring (Alternately use O-ring 12 x 2)	5	5
10	Label	1	1
11	Screw M3	4	4

* *Recommended Spares*



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**LEVER OPERATED
DIRECTION CONTROL VALVE
Model : DL06*****11**

Ref. No. D 09497
Release 03 / 2022

ENGINEERING - 1 of 4

Description

Completely encapsuiated mechanism for protection against dirt. Five chamber design for better reduction in dynamic force and longer valve life.

Valve mounting interface confirms to ISO 4401-03-02.

Mounting style - Subplate body. Available as spring centred, spring off-set or detented model.

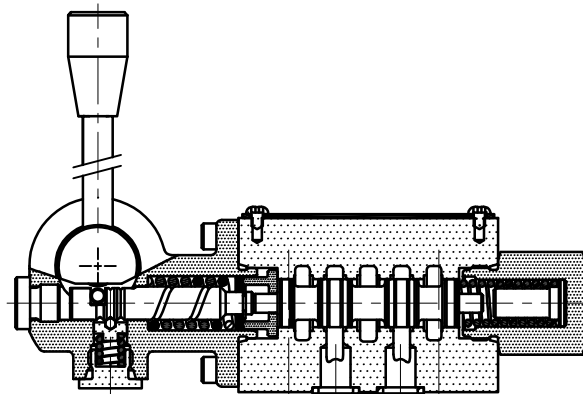
Operting Head can be rotated by 90° x 4 around spool axis for flexibility in mounting.

Hand lever can also be assembled, so that it is parallel to the axis at valve's netrual position.

All spool and bodies are interchangeable, simplifying maintence.

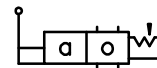


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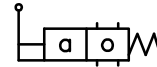


Hydraulic symbol

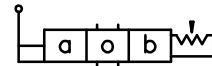
2 position detented



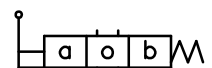
2 position spring offset



3 position detented



3 position spring centred



Technical specifications

Construction	:	Spool type.
Mounting / Standard	:	Subplate body. Interface as per ISO 4401 - 03 - 02
Mounting position	:	Optional, horizontal spool axis preferred.
Flow direction	:	As per spool type .
Maximum operating pressure	:	For port P, A, and B.....315 bar. (Standered valve) For port T.....100 bar. Pressure drop in the tank line adversely affect the returning speed of the cylinder, Hence must be kept as low as possible
Hydraulic medium	:	Mineral oil.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +70°C.
Fluid cleanliness requirement	:	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	:	refer performance curve.
Flow direction	:	Refer spool chart.
Mass (approx)	:	1.5 kg

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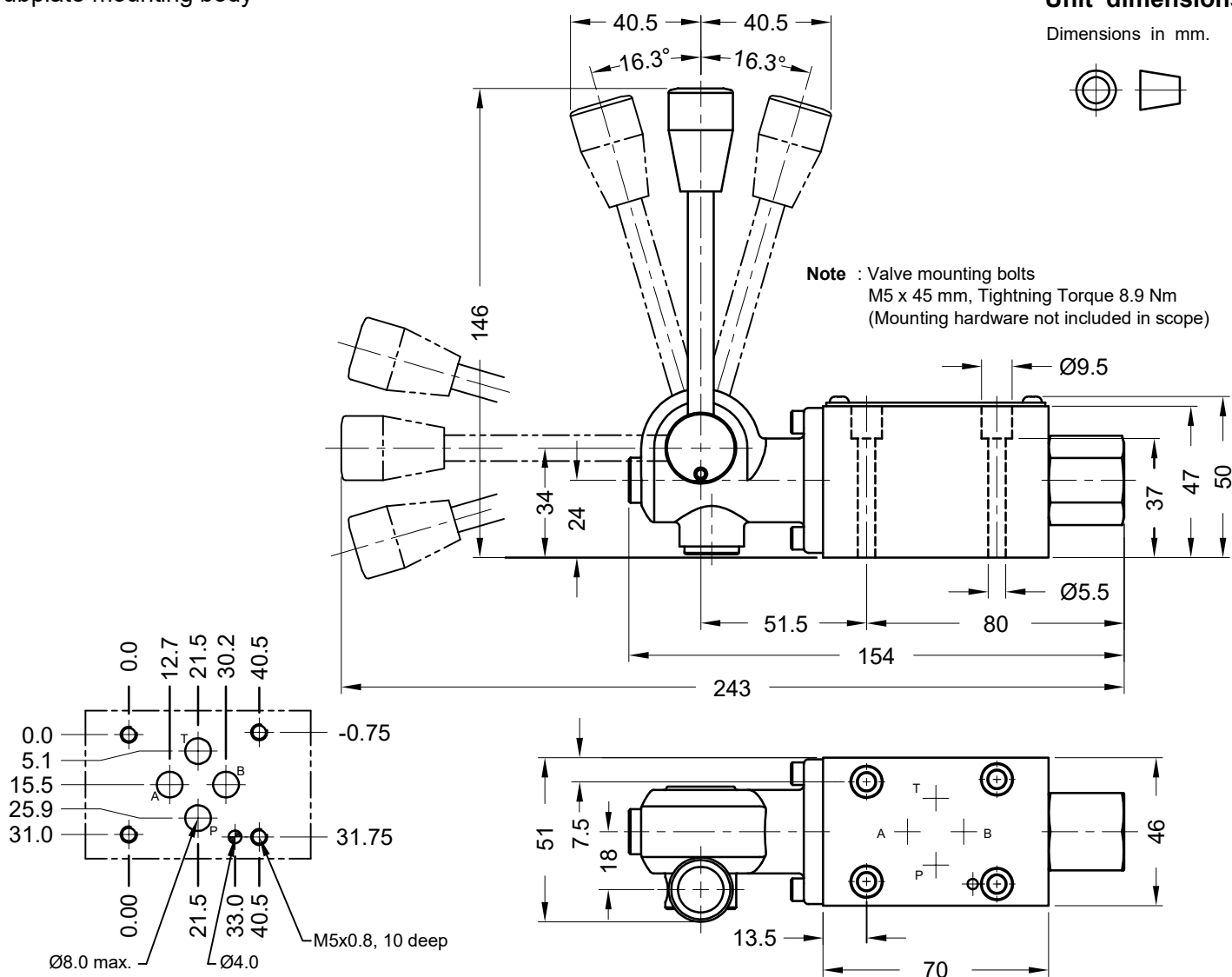
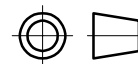
78-80, Machhe Industrial Estate,
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Phone : +91-(0)831- 2411001
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E-mail : polyhydon@gmail.com
Website : www.polyhydon.com

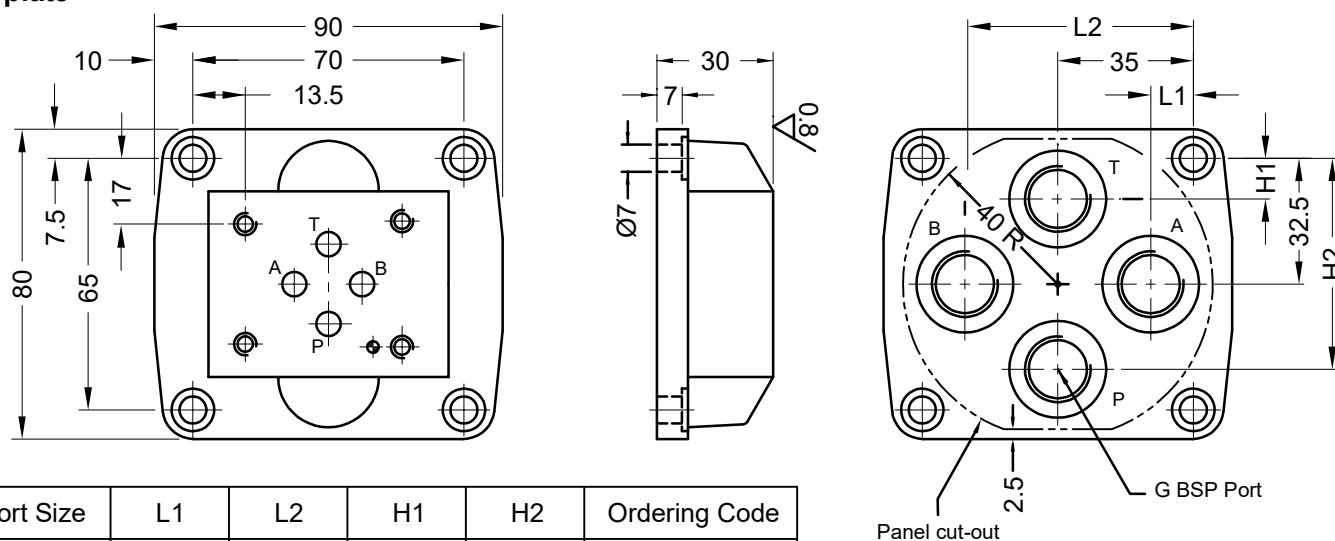
Subplate mounting body

Unit dimensions

Dimensions in mm.



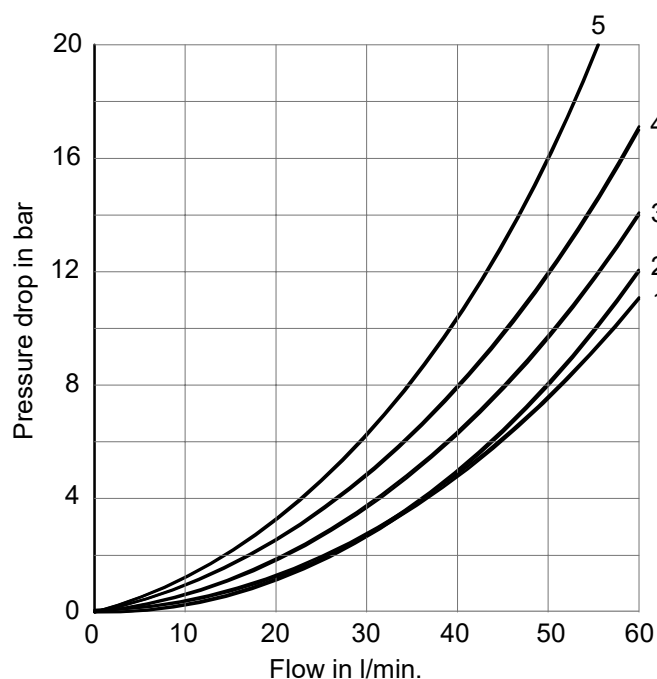
Subplate



Port Size	L1	L2	H1	H2	Ordering Code
G ¼	17	53	15.5	49.5	1 SD 03 G 02
G ⅜	11	59	10.5	54.5	1 SD 03 G 03

**Performance Curves for DL 06**

(Pressure drop related to flow, with Oil viscosity 46 cSt at 40°C)



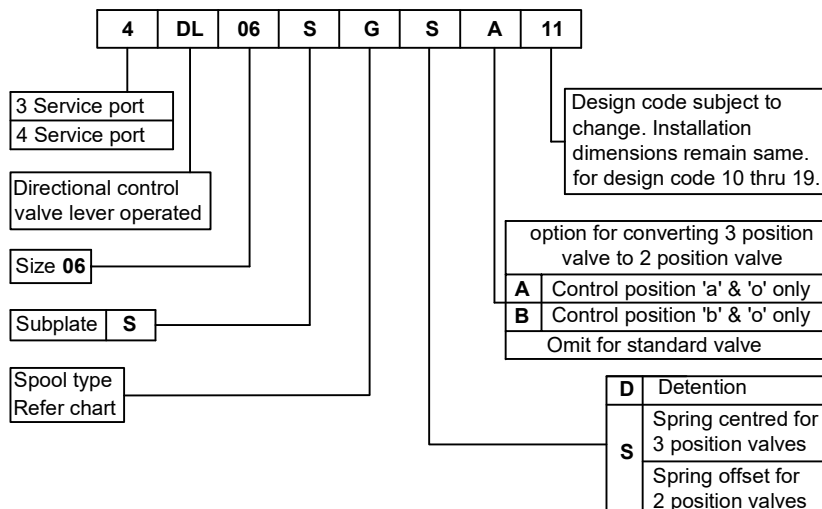
Spool type	Direction of flow / Curve no.				
	P to A	P to B	A to T	B to T	P to T
A	3	3	-	-	-
B	3	3	-	-	-
C	3	3	3	1	-
D	3	3	3	1	-
E	2	2	3	3	-
F	2	2	2	2	-
G	5	5	5	5	4
H	2	2	1	1	3
J	2	2	1	1	-
L	2	2	1	3	-
M	2	2	3	3	-
P	2	2	2	2	-

Spool Chart

Type	Symbol	Crossover	Type	Symbol	Crossover
	a 0 b			a 0 b	
	A B			A B	
A			J		
B			L		
C			M		
D			P		
E			Q		
F			U		
G			V		
H					



Ordering code



Note :

Sub-plate to be ordered separately.



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**SOLENOID OPERATED
DIRECTION CONTROL VALVE**
Model : DE06****07
(CETOP 03)

Ref. No. D 09321
Release 01 / 2026

ENGINEERING - 1 of 4

Description

Direct solenoid operated, spool type directional control valve with completely encapsulated mechanism for protection against dirt. Five chamber design for better reduction in dynamic force and longer valve life.

Valve mounting interface confirms to ISO 4401-03-02.

Mounting style - Subplate body. Available as spring centred, spring off-set or detented model.

All spool and bodies are interchangeable, simplifying maintenance.

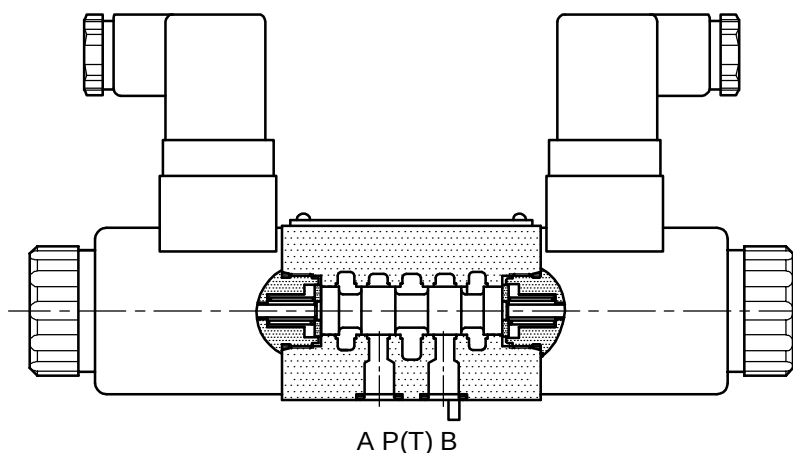
Wet pin DC solenoid ensure better heat dissipation, quieter operation, thereby increasing solenoid life.

Moulded solenoid coil eliminate moisture problem and ingress of dust.

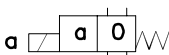
Plug-in-coil for ease of maintenance. Indicator light are a standard feature.

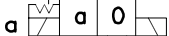


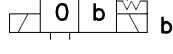
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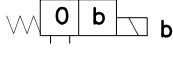


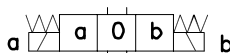
Hydraulic symbol

2 Position spring return 

2 Position detent 

* 2 Position detent
(For spool 'B' only) 

* 2 Position spring return
(For spool 'B' only) 

3 Position spring return 

Technical specifications

Electrical :

D.C. Voltage	:	12V. and 24V.
A.C. Voltage	:	110 V. and 220V at 50Hz.
Power requirement	:	30W
Duty cycle	:	Continuous
Switching time (ON)	:	40 ms
(OFF)	:	30ms
Max. ambient temperature	:	50°C
Max. coil temperature	:	100°C
Max. switching frequency	:	15000 / hour.
Protection	:	IP65.

Note: AC solenoid coil is basically a DC coil with built-in rectifier.

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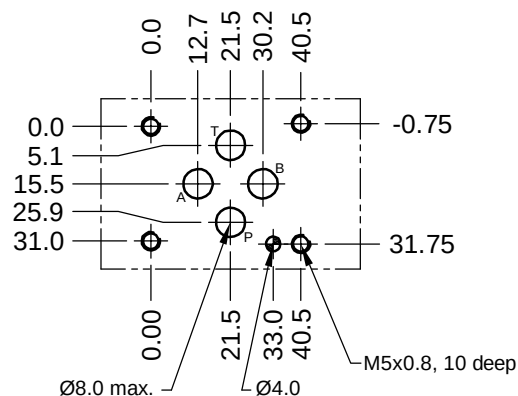
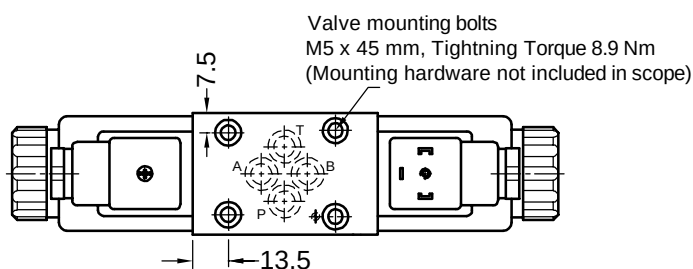
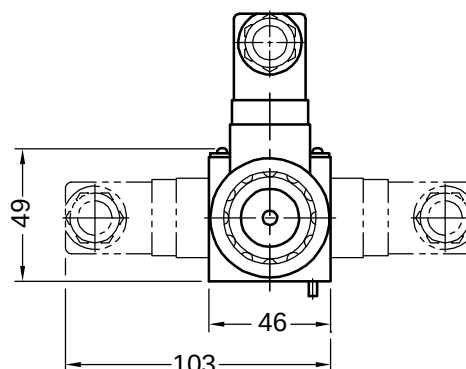
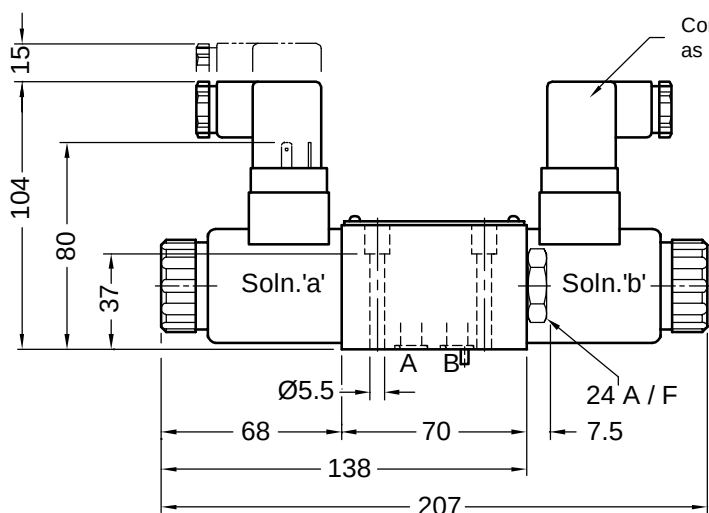
78-80, Machhe Industrial Estate,
Machhe, Belgaum - 590 014. INDIA.

Phone : +91-(0)831- 2411001
E-mail : sales@polyhydron.com
Website : www.polyhydron.com



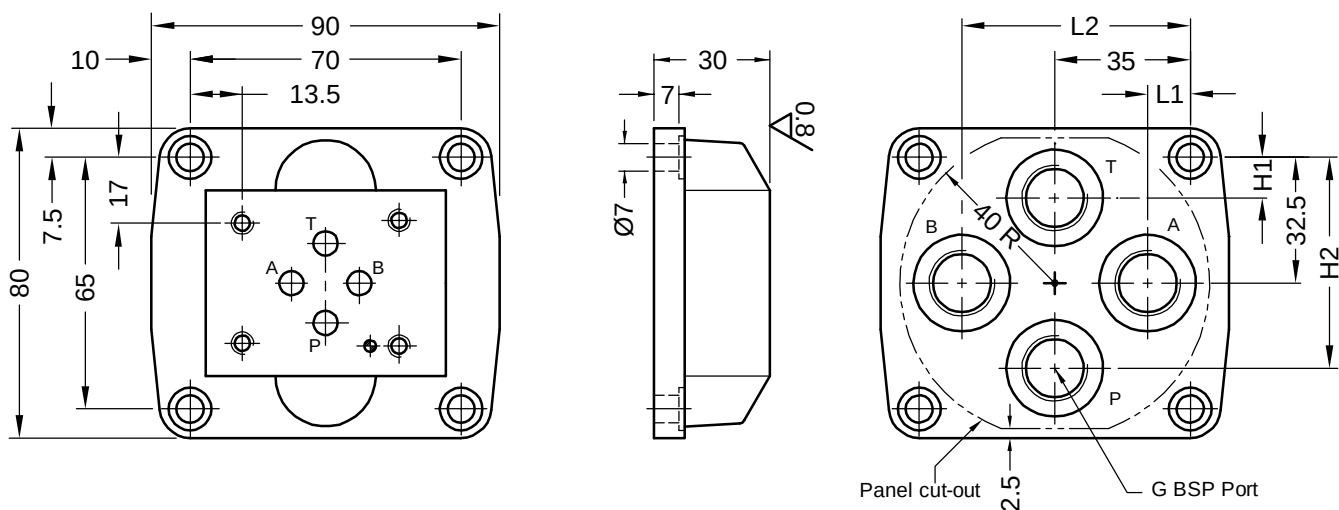
Construction	:	Spool type.
Mounting / Standard	:	Subplate body.
		Interface as per ISO 4401 - 03 - 02
Mounting position	:	Optional, horizontal spool axis preferred.
Flow direction	:	As per spool type .
Maximum operating pressure	:	For port P, A, and B.....315 bar. (Standarded valve)
		For port T.....100 bar.
		Pressure drop in the tank line adversely affect the returning speed of the cylinder, Hence must be kept as low as possible (For spool A & B type, port T to be used as drain line, if the operating pressure exceeds 150 bar.)
Hydraulic medium	:	Mineral oil.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +80°C.
Fluid cleanliness requirement	:	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	:	refer performance curve.
Flow direction	:	Refer spool chart.
Mass (approx)	:	valve with single solenoid - 1.2 kg.
		valve with Double solenoid - 1.8 kg.

Dimensions in mm.





Subplate

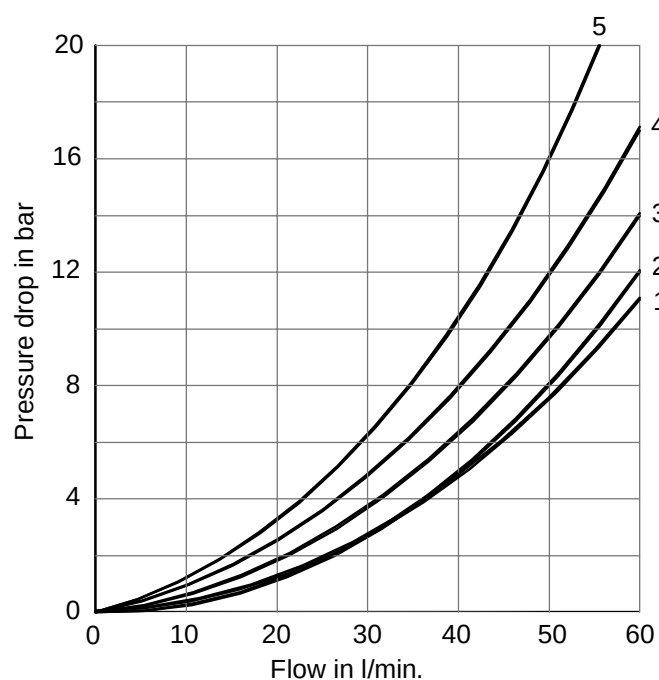


Port Size	L1	L2	H1	H2	Ordering Code
G 1/4	17	53	15.5	49.5	1 SD 03 G 02
G 3/8	11	59	10.5	54.5	1 SD 03 G 03

Note : Sub-plate to be ordered separately.

Performance Curves for DE 06

(Pressure drop related to flow, with Oil viscosity 46 cSt at 40°C)



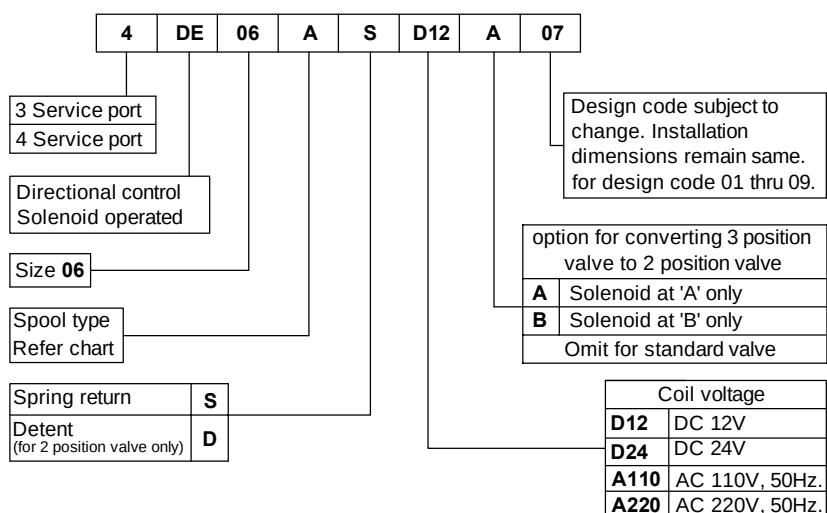
Spool type	Direction of flow / Curve no.				
	P to A	P to B	A to T	B to T	P to T
A	3	3	-	-	-
B	3	3	-	-	-
C	3	3	3	1	-
D	3	3	3	1	-
E	2	2	3	3	-
F	2	2	2	2	-
G	5	5	5	5	4
H	2	2	1	1	3
J	2	2	1	1	-
L	2	2	1	3	-
M	2	2	3	3	-
P	2	2	2	2	-



Spool Chart

Type	Symbol	Crossover	Type	Symbol	Crossover
	a 0 b A B P T	A B P T		a 0 b A B P T	A B P T
A			J		
B			L		
C			M		
D			P		
E			Q		
F			U		
G			V		
H					

Ordering code



Note :

Ordering code for seal kit : 4DE06-SLKT.



SOLENOID OPERATED DIRECTION CONTROL VALVE DES *

Ref. No.D06631
Release : 09 / 2025

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ENGINEERING - 1 of 4

Description

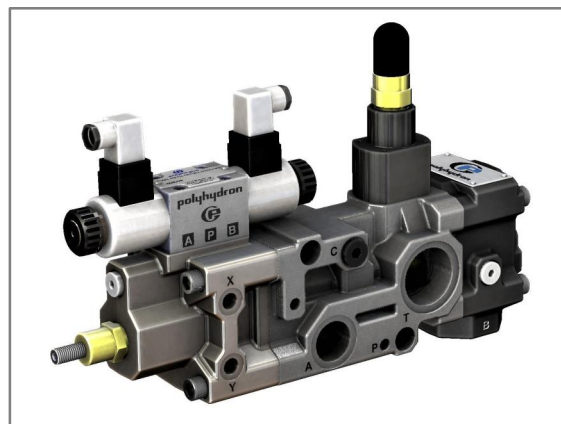
Designed for operating single acting up-stroking cylinders, which are expected to hold pressure during their working cycle.

The valve also ensure smooth decompression prior to return of the cylinder.

Large return oil passage in the valve helps to return of the cylinder rapidly.

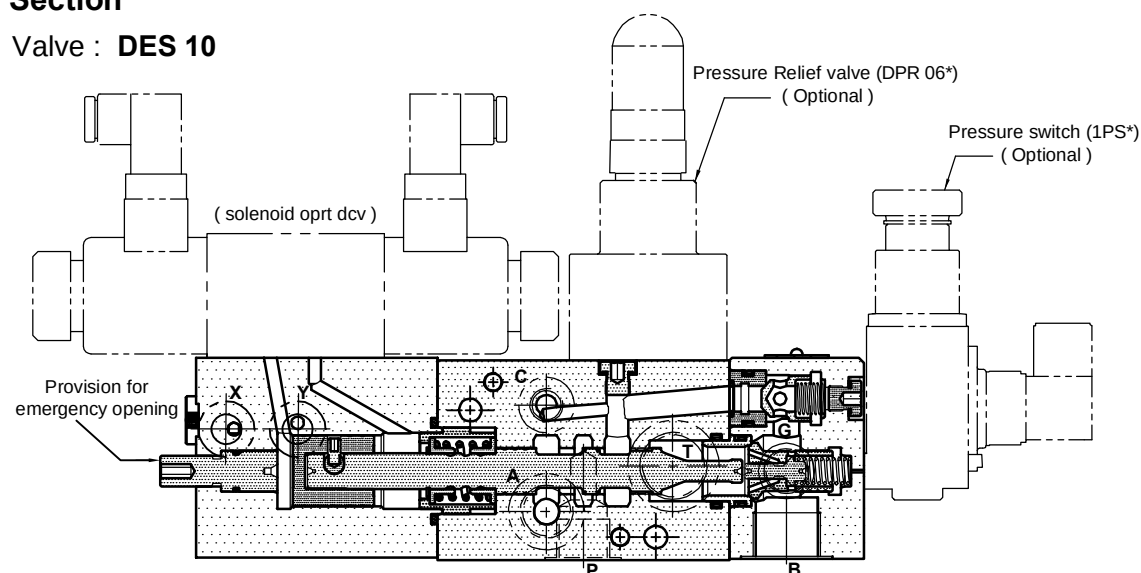
Operations of multiple independent single acting cylinders can be achieved by using the valves connected in series. (Please refer circuit on last page.)

The valves can be supplied with features for mounting Pressure Regulator valve and Pressure Switch to eliminate interconnecting pipes.

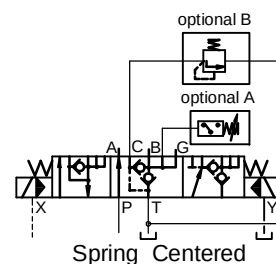


Section

Valve : **DES 10**



Hydraulic symbol



Technical specifications

Construction	Spool with Poppet.
Mounting / Standard	Threaded body (Company standard)
Mounting position	Optional, horizontal spool axis preferred.
Flow direction	As per Hydraulic Symbol.
Maximum operating pressure	At port P, A, B, C, G.....350 bar. Port T.....Pressure drop in the tank line adversely affect the returning speed of the cylinder, Hence must be kept as low as possible.
Pilot pressure	At port X= (Cylinder Pr. /16) + 1 bar.
Pilot oil flow	2 l/min (min).
Hydraulic medium	Mineral oil.
Viscosity range	10 cSt to 380 cSt
Fluid temperature range	-20°C to +70°C.
Fluid cleanliness requirement	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	refer performance curve.
Mass (approx)	DES 10.....10.5 kg. DES 2019.5 kg.

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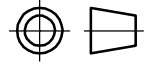
78-80, Machhe Industrial Estate,
Machhe, Belgaum - 590 014. INDIA.

Phone : +91-(0)831- 2411001
Fax : +91-(0)831- 2411002
E-mail : polyhydron@gmail.com
Website : www.polyhydron.com

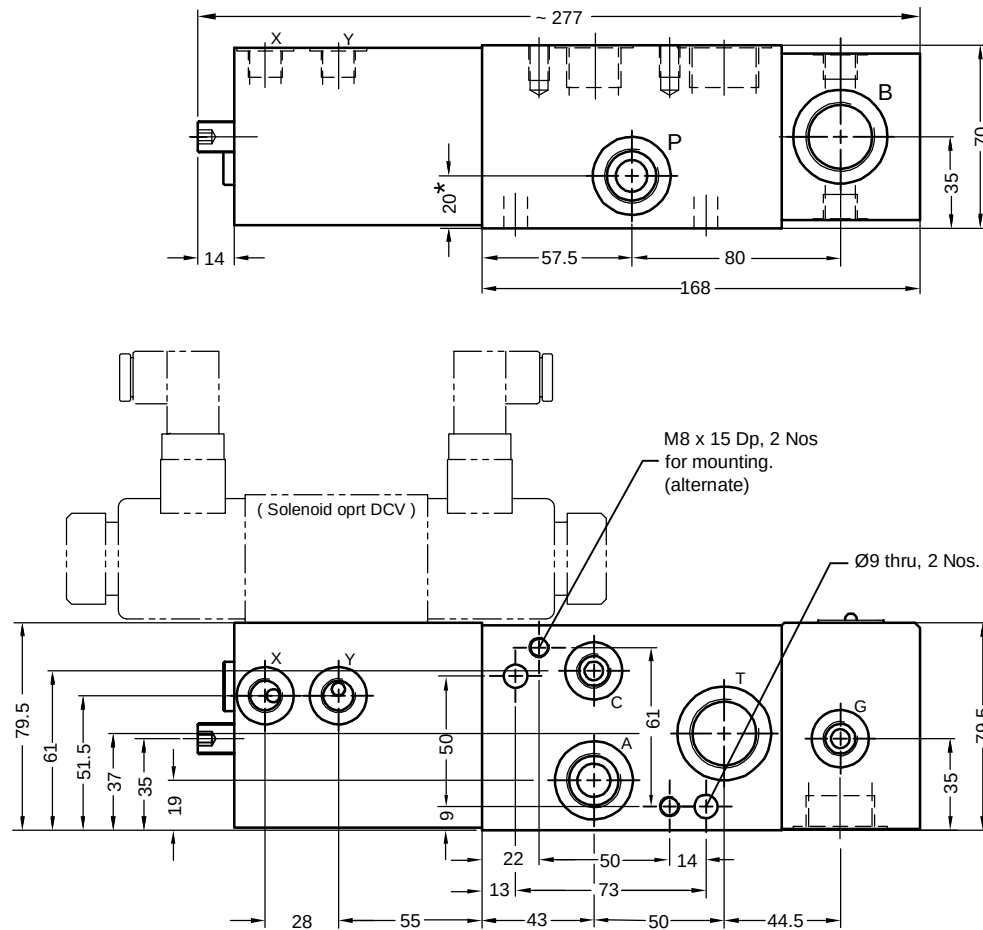
Unit dimensions

Valve model : **DES 10**

Dimensions in mm.



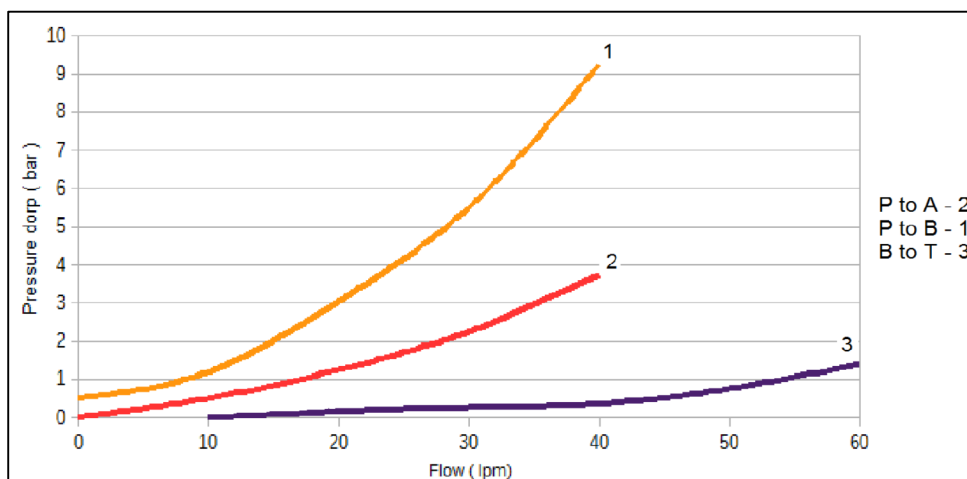
Port size details	
Port	Size
A , P	G1/2
B , T	G3/4
G , C X , Y	G1/4



Note : Valve with Mirror image face option, will have ports A, C, T and G on opposite face and dimension 20 (marked *) of port P will be 50. (Ask for details, if needed)

Performance Curves for DES 10

(Pressure drop related to flow, with 46 cSt at 40°C)

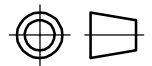




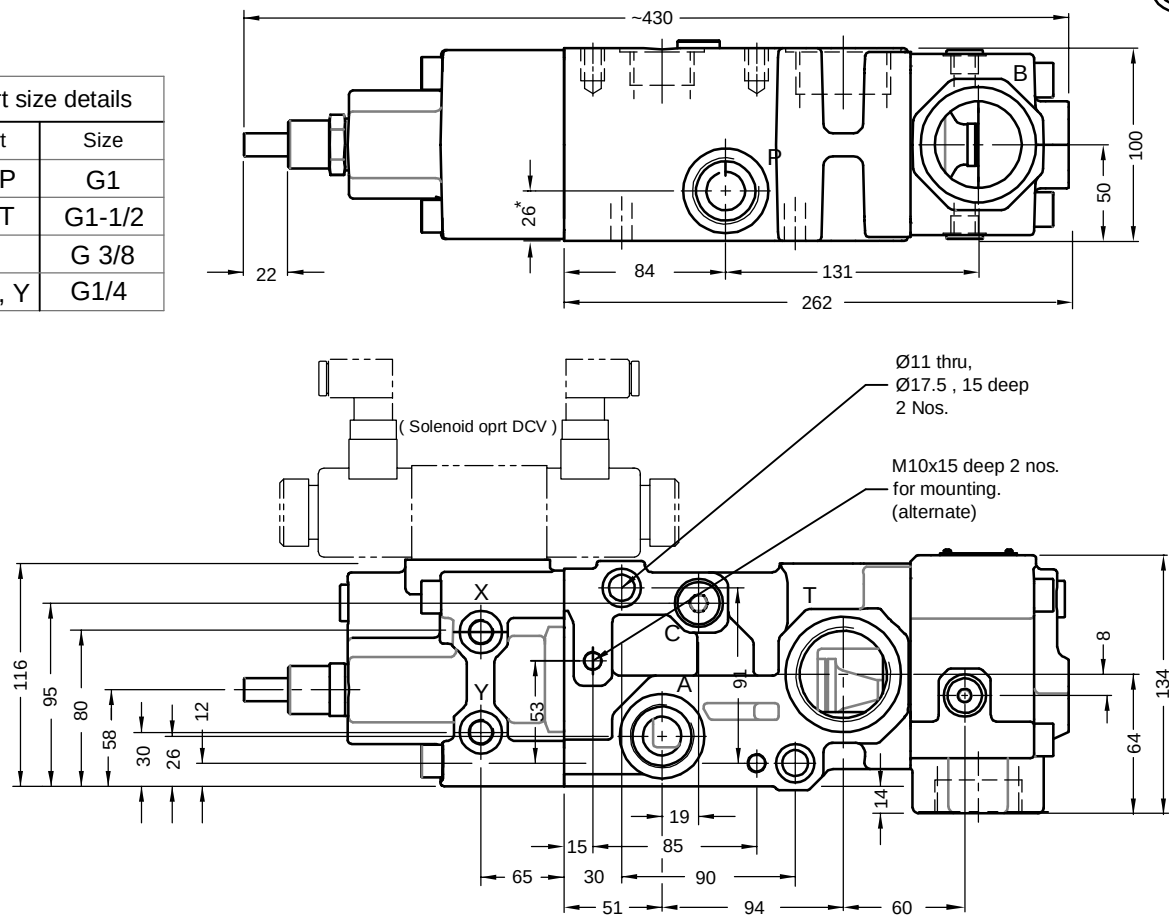
Unit dimensions

Valve model : **DES 20**

Dimensions in mm.



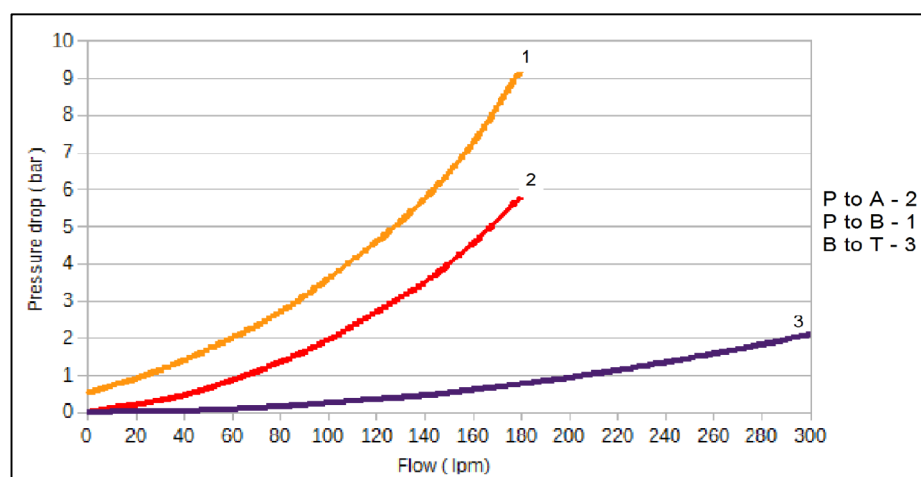
Port size details	
Port	Size
A , P	G1
B , T	G1-1/2
C	G 3/8
G, X, Y	G1/4



Note : Valve with Mirror image face option, will have ports A, C, T and G on opposite face and dimension 26 (marked *) of port P will be 74. (Ask for details, if needed)

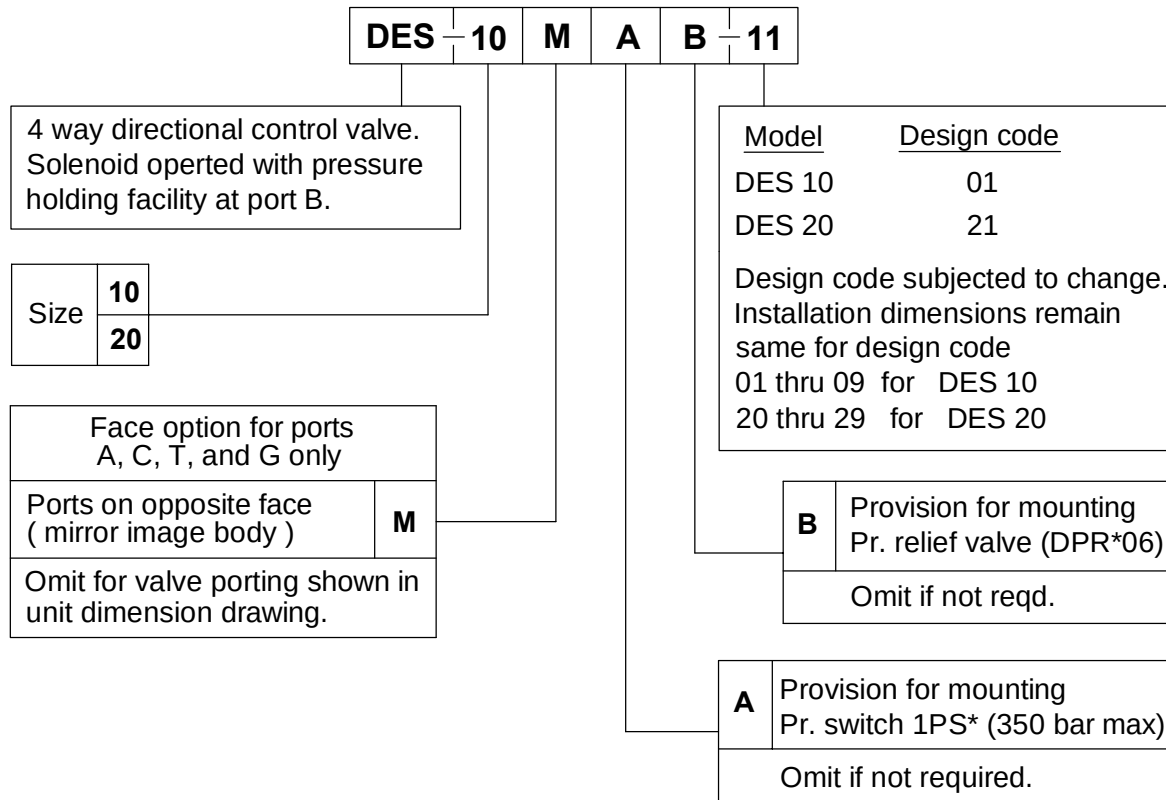
Performance Curves for DES 20

(Pressure drop related to flow, with 46 cSt at 40°C)



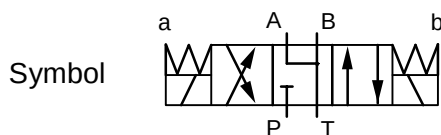


Ordering code

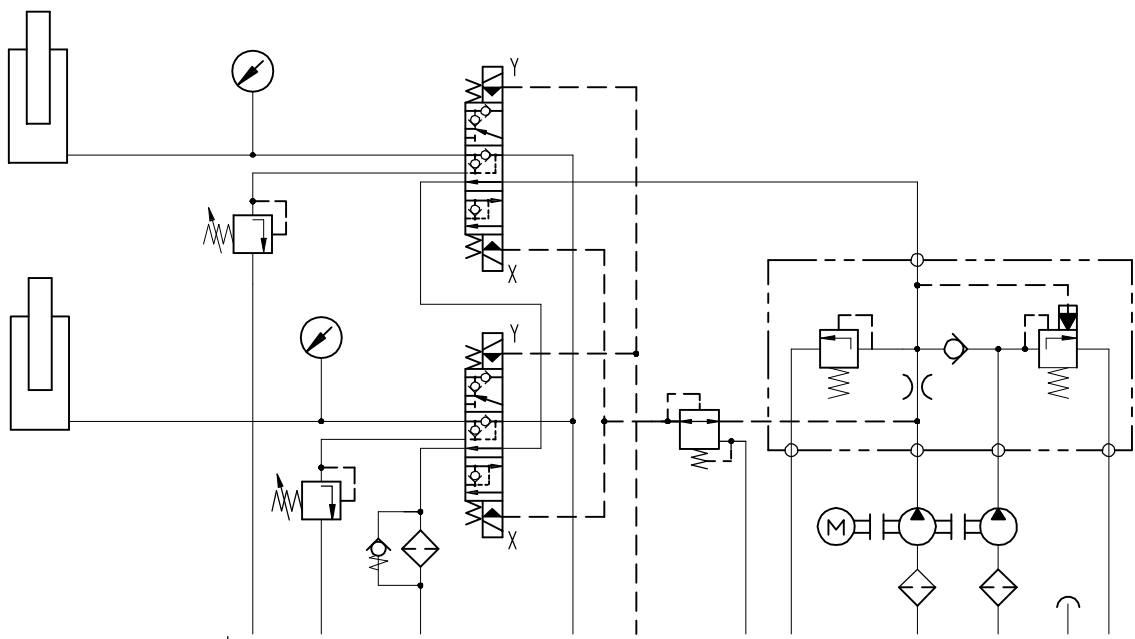


NOTE : For actuation of this valve, solenoid DCV (refer spool type) to be ordered separately.

Spool type required : 'J' type



Hydraulic circuit showing typical application of valve DES10 / DES 20





A Polyhydon Group Company

HAND LEVER OPERATED DIRECTION CONTROL VALVE DLS *

Ref. No.D09072
Release : 09 / 2025
ENGINEERING - 1 of 4

Description

Designed for operating single acting up-stroking cylinders, which are expected to hold pressure during their working cycle.

The valve also ensure smooth decompression prior to return of the cylinder.

Large return oil passage in the valve helps to return of the cylinder rapidly.

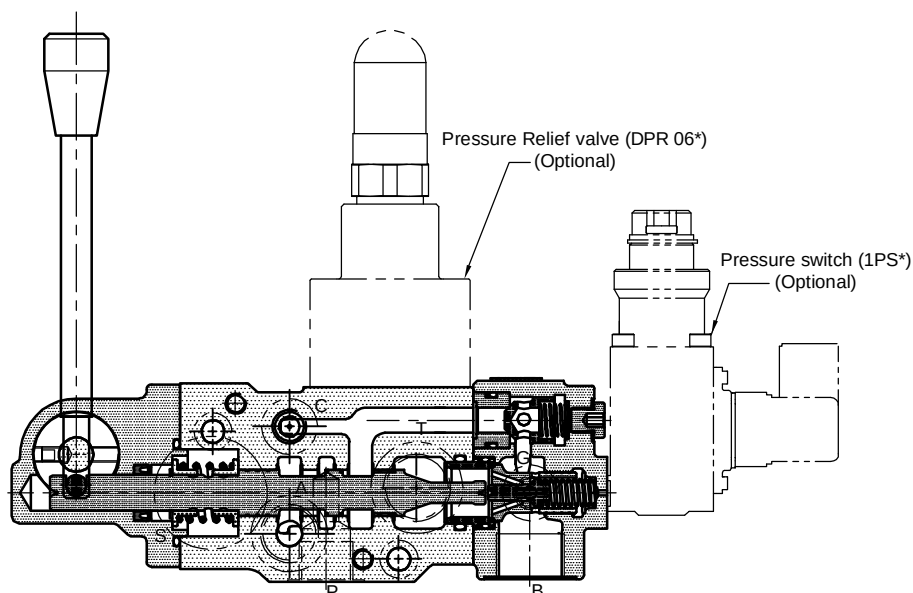
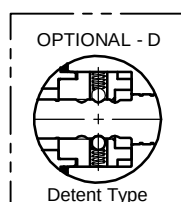
Operations of multiple independent single acting cylinders can be achieved by using the valves connected in series. (Please refer circuit on last page.)

The valves can be supplied with features for mounting Pressure Regulator valve and Pressure Switch to eliminate interconnecting pipes.

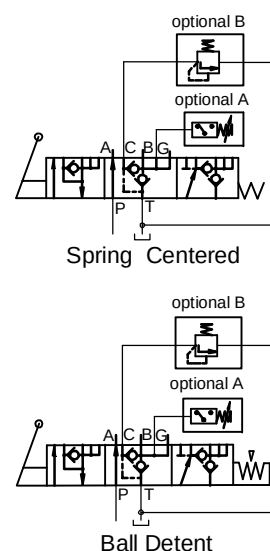


Section

Valve : **DLS 10**



Hydraulic symbol



Technical specifications

Construction	Spool with Poppet.
Mounting / Standard	Threaded body (Company standard)
Mounting position	Optional, horizontal spool axis preferred.
Flow direction	As per Hydraulic Symbol.
Maximum operating pressure.....	At port P, A, B, C, G.....350 bar Port T.....Pressure drop in the tank line adversely affect the returning speed of the cylinder, Hence must be kept as low as possible
Hydraulic medium	Mineral oil.
Viscosity range	10 cSt to 380 cSt
Fluid temperature range	-20°C to +70°C
Fluid cleanliness requirement	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	refer performance curve.
Mass (approx)	DLS 10.....8 kg DLS 2018 kg

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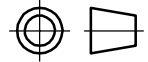
Phone : +91-(0)831- 2411001
Fax : +91-(0)831- 2411002
E-mail : polyhydon@gmail.com
Website : www.polyhydon.com



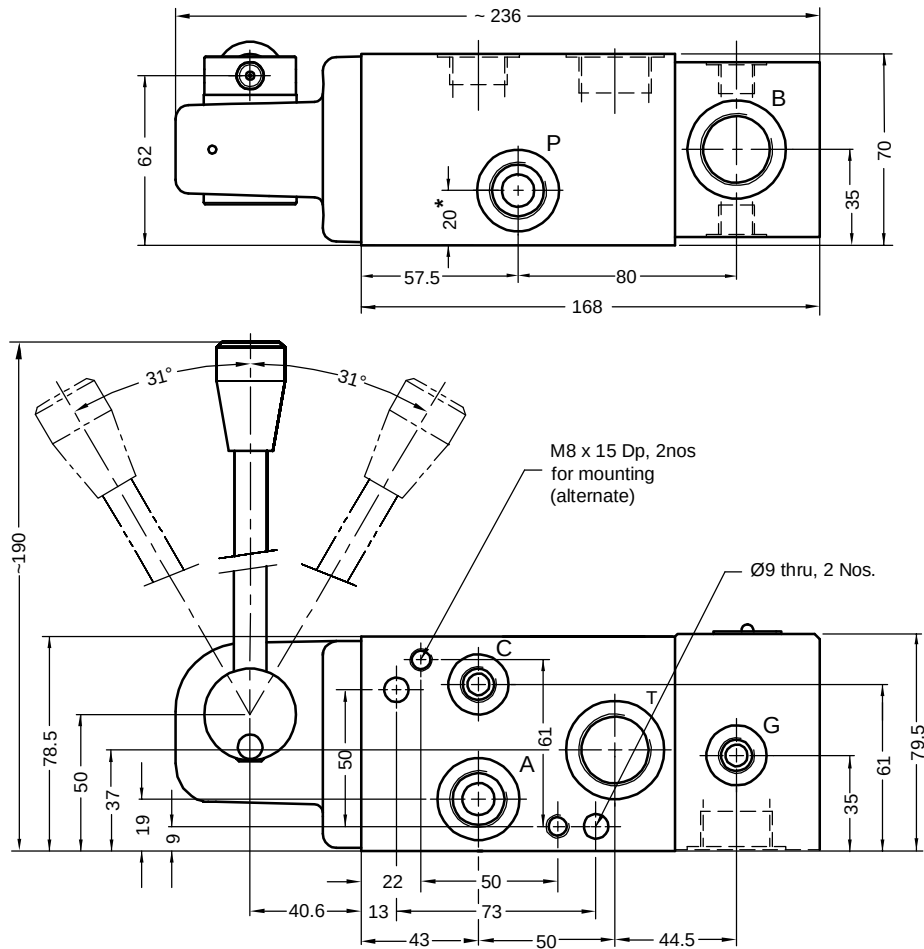
Unit dimensions

Valve model : **DLS 10**

Dimensions in mm.

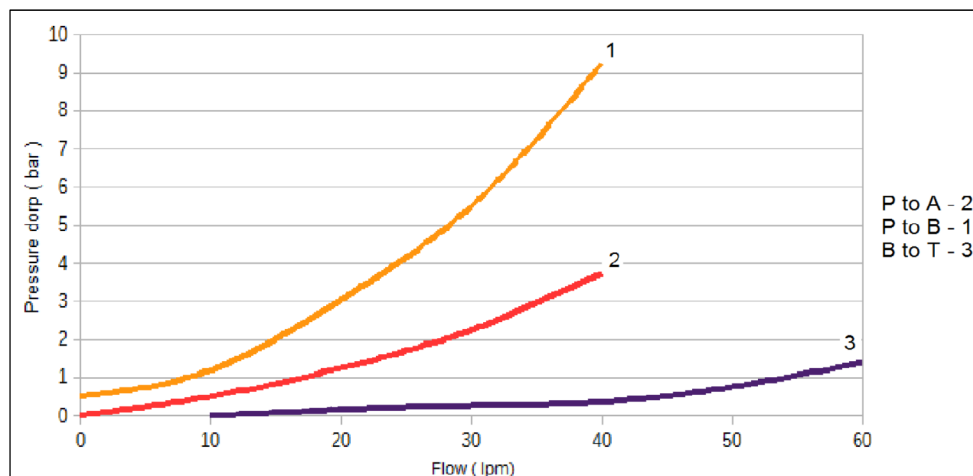


Port size details	
Port	Size
A , P	G1/2
B , T	G3/4
C , G	G1/4



Note : Valve with Mirror image face option, will have ports A, C, T and G on opposite face and dimension 20 (marked *) of port P will be 50. (Ask for details, if needed)

Performance Curves for DLS 10 (Pressure drop related to flow, with 46 cSt at 40°C)



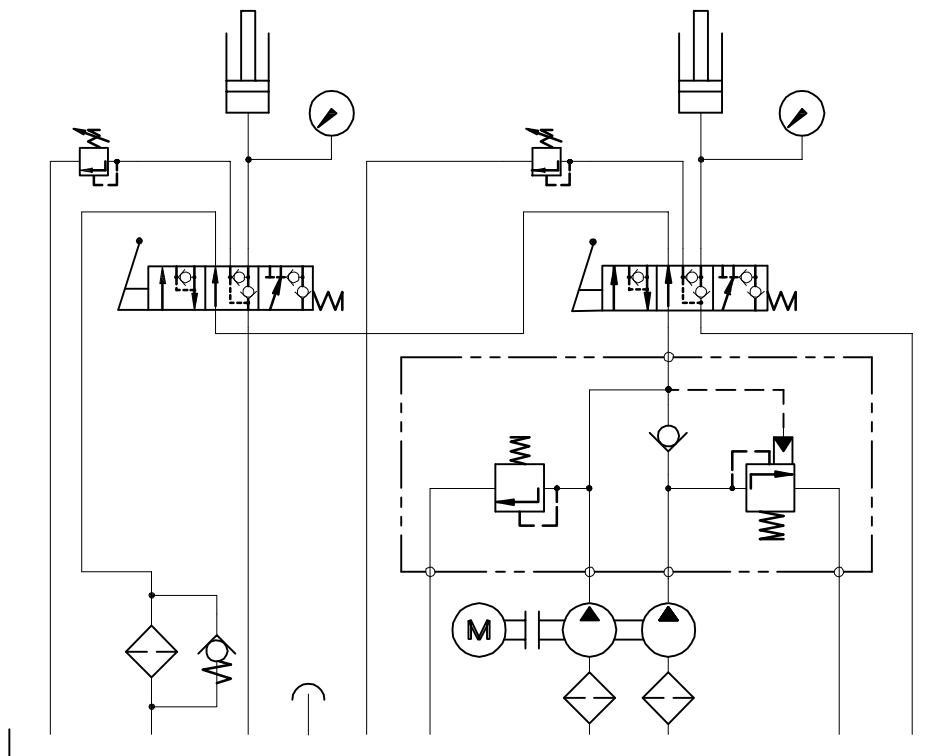


Ordering code

DLS		10	S	M	A	B	11						
4 way directional control valve Hand lever operated with pressure holding facility at port B.													
Size	10												
	20												
Spring Return	S												
Detent	D												
Face option for ports A, C, T, and G only													
Ports on opposite face (mirror image body)						M							
Omit for valve porting shown in unit dimension drawing.													
<table border="1"><thead><tr><th>Model</th><th>Design code</th></tr></thead><tbody><tr><td>DLS 10</td><td>11</td></tr><tr><td>DLS 20</td><td>21</td></tr></tbody></table>								Model	Design code	DLS 10	11	DLS 20	21
Model	Design code												
DLS 10	11												
DLS 20	21												
Design code subjected to change. Installation dimensions remain same for design code 10 thru 19 for DES 10 20 thru 29 for DES 20													
						B	Provision for mounting Pr. relief valve (DPR*06)						
Omit if not required.													
						A	Provision for mounting Pr. switch 1PS* (350 bar max)						
Omit if not required.													

Note : Pressure switch (1PS*) and Pr. relief valve (DPR*06) to be ordered separately.

Hydraulic circuit showing typical application of valve DLS10 / DLS 20





A Polyhydron Group Company

PILOT OPERATED DIRECTION CONTROL VALVE Model : 4DP10*S*10

Ref. No. D32200
Release 05 / 2022

ENGINEERING - 1 of 4

Description

Completely encapsulated mechanism for protection against dirt. Five chamber design for better reduction in dynamic force and longer valve life.

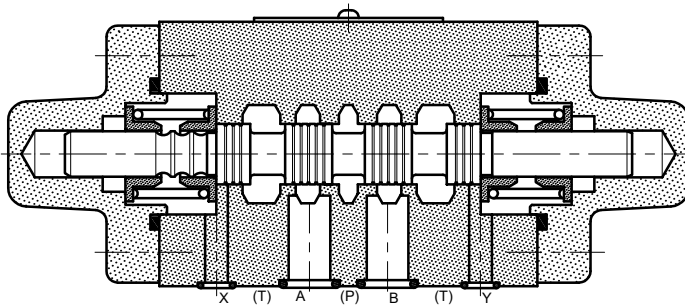
Mounting style - Subplate body. Available as spring centred and spring off-set position

Valve mounting interface confirms to international and national standards.

All spool and bodies are interchangeable, simplifying maintenance.

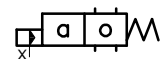


Section

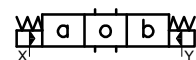


Hydraulic symbol

2 position spring offset



3 position spring centred



Technical specifications

Construction	:	Spool type.
Mounting / Standard	:	Subplate body. Interface as per ISO 4401 - AC - 05 - 4 - A and IS 10187 -10 mm diameter nominal port.
Mounting position	:	Optional, horizontal spool axis preferred.
Flow direction	:	As per spool type .
Maximum operating pressure	:	For port P, A, and B.....350 bar. (Standered valve) For port P, A, and B.....700 bar. (High Pressure valve) For port X, Y.....100 bar. For port T.....100 bar. Pressure drop in the tank line adversely affect the returning speed of the cylinder, Hence must be kept as low as possible
Minimum Pilot pressure	:	5 bar.
Hydraulic medium	:	Mineral oil.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +70°C.
Fluid cleanliness requirement	:	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	:	Refer performance curve.
Mass (approx)	:	5 kg

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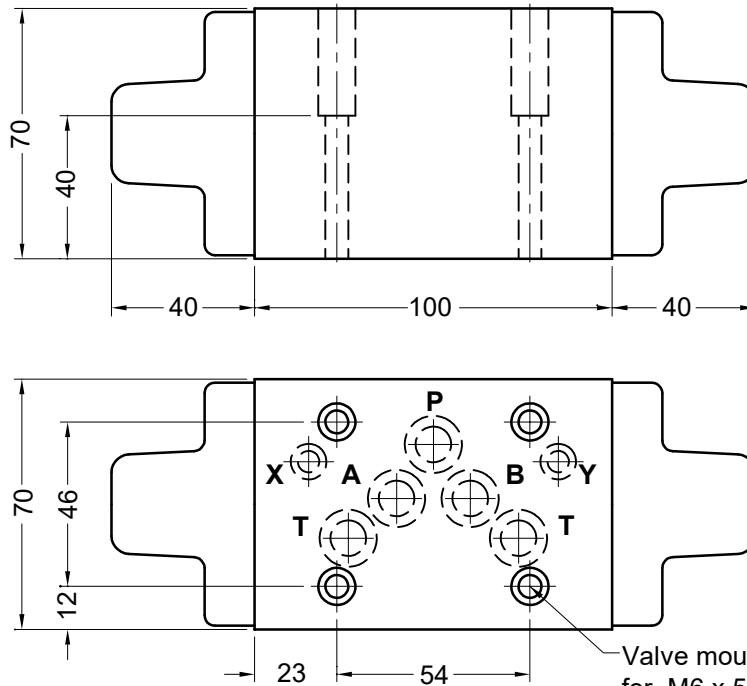
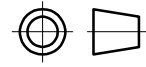
Phone : +91-(0)831- 2411001
Fax : +91-(0)831- 2411002
E-mail : polyhydron@gmail.com
Website : www.polyhydron.com



Subplate mounting body

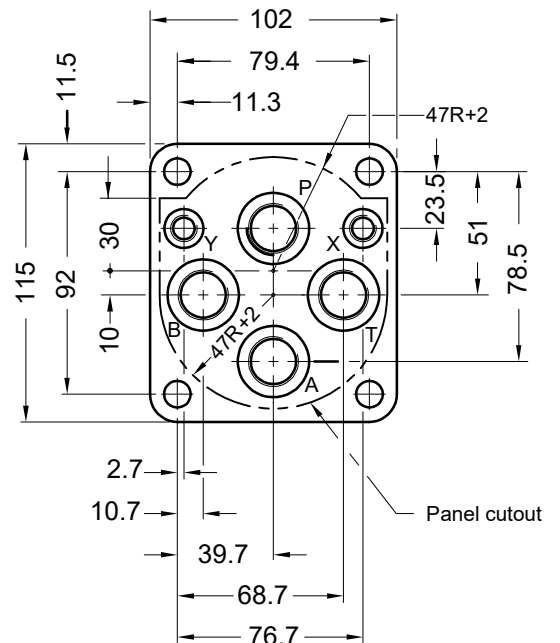
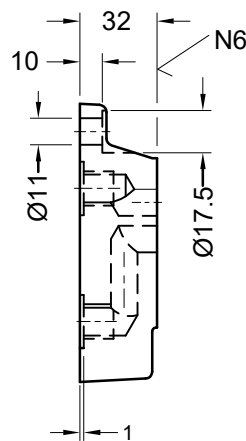
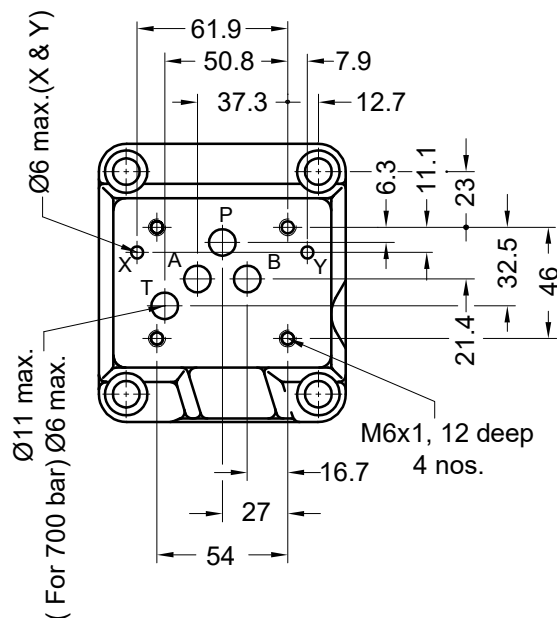
Unit dimensions

Dimensions in mm.



Valve mounting Clearance hole,
for M6 x 50L S.H. Capscrew. 4 Nos.
property class 10.9, tightening torque 16 Nm.

Subplate



Subplate Ordering Code	Port A, B, P, T	Port X, Y	Approx Mass	Sub-plate Fixing Screws	Tightening Torque
G1 03 XY	G 1/2"	G 1/8"	2.5 kg.	M10 x 25 L	70 N-m

Note

Sub-plate to be ordered separately.



Performance Curves for 4DP 10

(Pressure drop related to flow, with oil viscosity 46 cSt at 40°C)

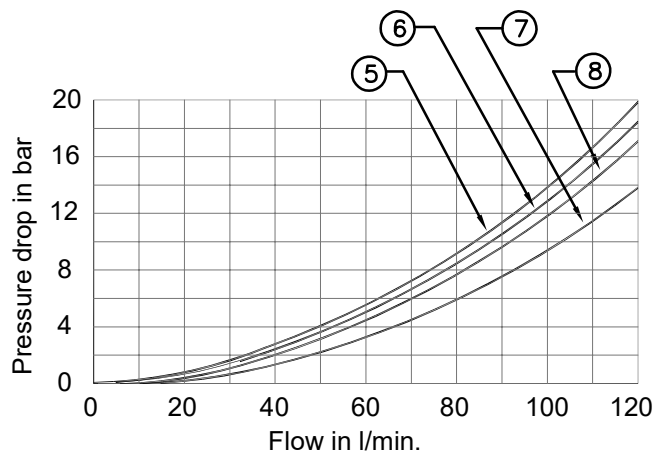
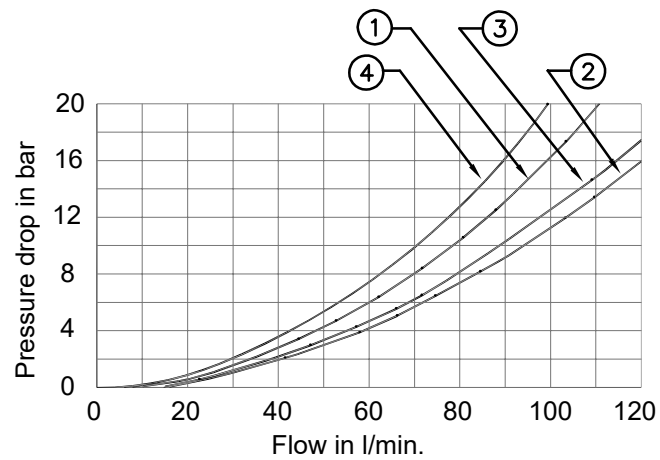
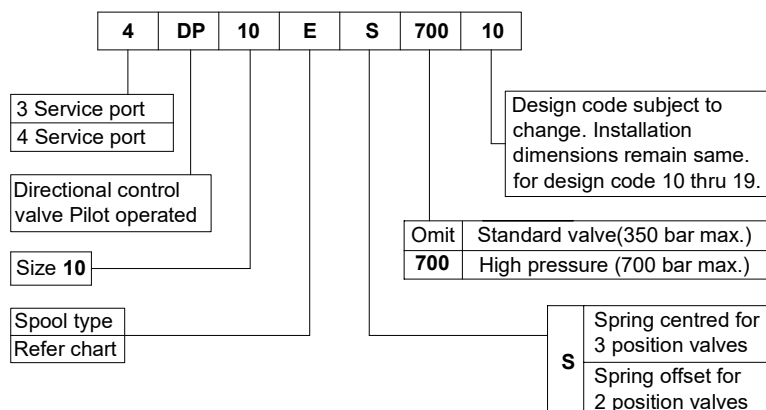


Table showing the relation between the spool type, direction of flow and curve graph to be referred to.

Spool type	Direction of flow / Curve no.				
	P to T	P to A	P to B	A to T	B to T
A	-	1	1	-	-
C	-	1	1	2	3
D	-	1	1	2	3
E	-	1	1	2	3
F	4	1	1	7	3
G	4	1	1	2	3
H	5	6	6	7	8
J	-	1	1	7	8
L	-	1	1	7	3
M	-	6	6	2	3
P	4	1	1	2	8
Q	-	1	1	2	3
U	-	1	1	7	8
W	-	1	1	2	3



Ordering code



Note

Ordering code for a complete Seal Kit : **DP*SLKT**

Spool Chart

Type	Symbol	Crossover	Type	Symbol	Crossover
	a 0 b			a 0 b	
	A B			A B	
A			Q		
	P T			P T	
C			U		
D			W		
E					
F					
G					
H					
J					
L					
M					
P					



A Polyhydron Group Company

LEVER OPERATED DIRECTION CONTROL VALVE Model : 4DL10*****10

Ref. No. D04909
Release 02 / 2024

ENGINEERING - 1 of 5

Description

Completely encapsuiated mechanism for protection against dirt. Five chamber design for better reduction in dynamic force and longer valve life.

Valve mounting interface confirms to National and International standards.

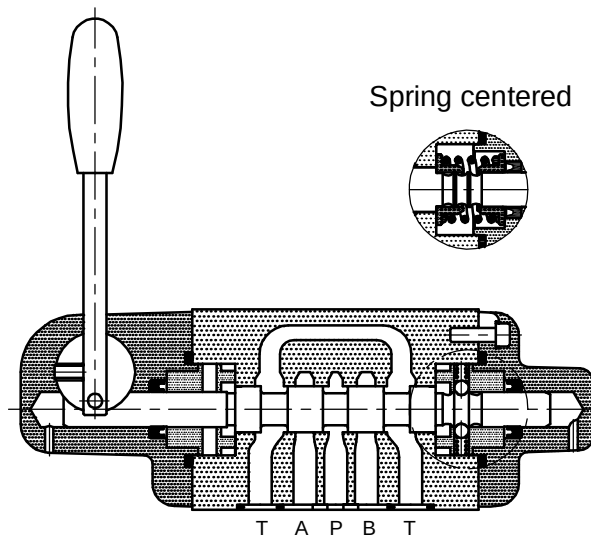
Mounting style - Subplate & Threaded body. Available as spring centered, spring off-set or detented model.

Operating Head can be rotated by 90° x 4 around spool axis for flexibility in mounting.

All spool and bodies are interchangeable, simplifying maintenance.

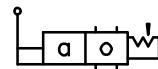


Section

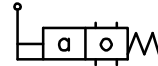


Hydraulic symbol

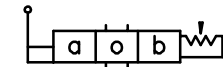
2 position detented



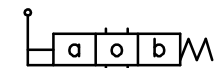
2 position spring offset



3 position detented



3 position spring centred



Technical specifications

Construction	:	Spool type.
Mounting / Standard	:	Subplate & Threaded body. Subplate body Interface as per ISO 4401 - AC - 05 - 4 - A and IS 10187 -10 mm diameter nominal port.
Mounting position	:	Optional, horizontal spool axis preferred.
Flow direction	:	As per spool type .
Maximum operating pressure	:	For port P, A, and B.....350 bar. (Standered valve) For port P, A, and B.....700 bar. (High Pressure valve) For port T.....100 bar. Pressure drop in the tank line adversely affect the returning speed of the cylinder, Hence must be kept as low as possible
Hydraulic medium	:	Mineral oil.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +70°C.
Fluid cleanliness requirement	:	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	:	Refer performance curve.
Mass (approx)	:	5.6 kg

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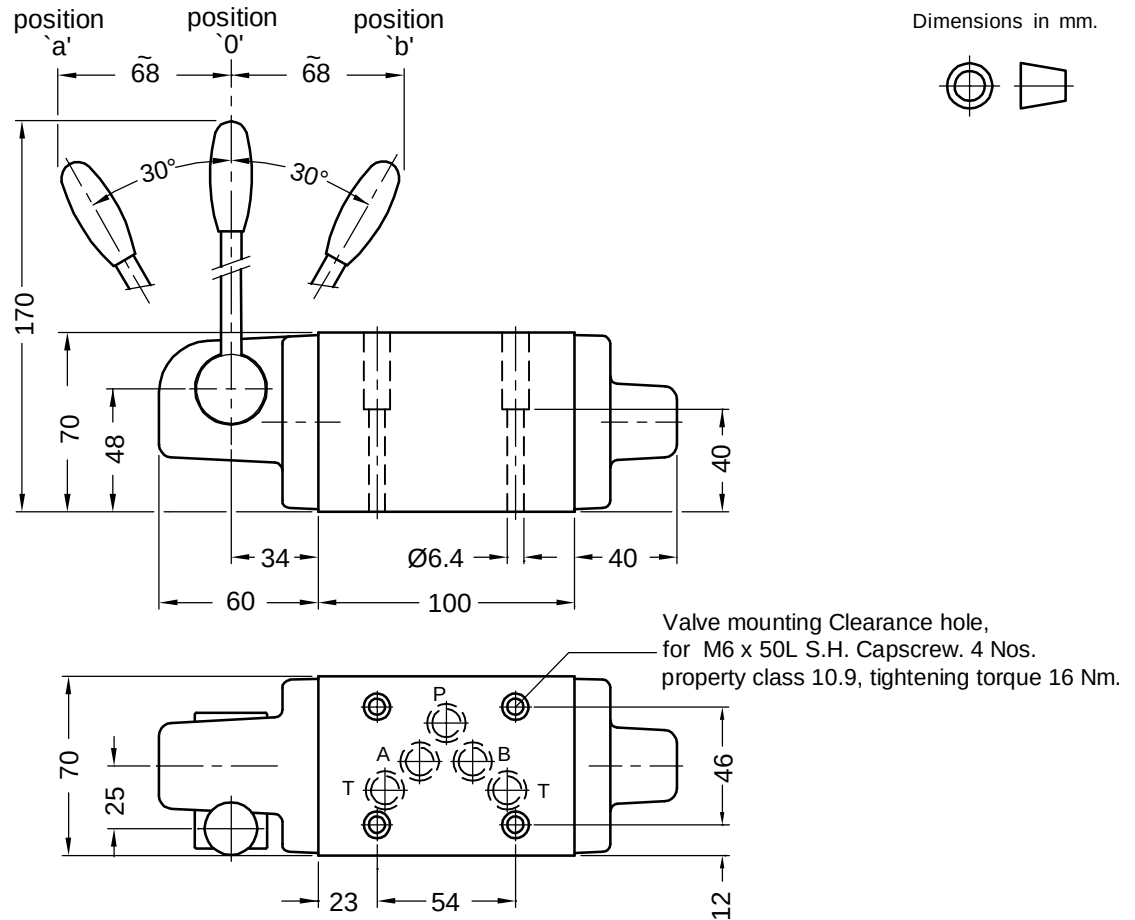
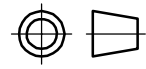
Phone : +91-(0)831- 2411001
Fax : +91-(0)831- 2411002
E-mail : sales@polyhydron.com
Website : www.polyhydron.com



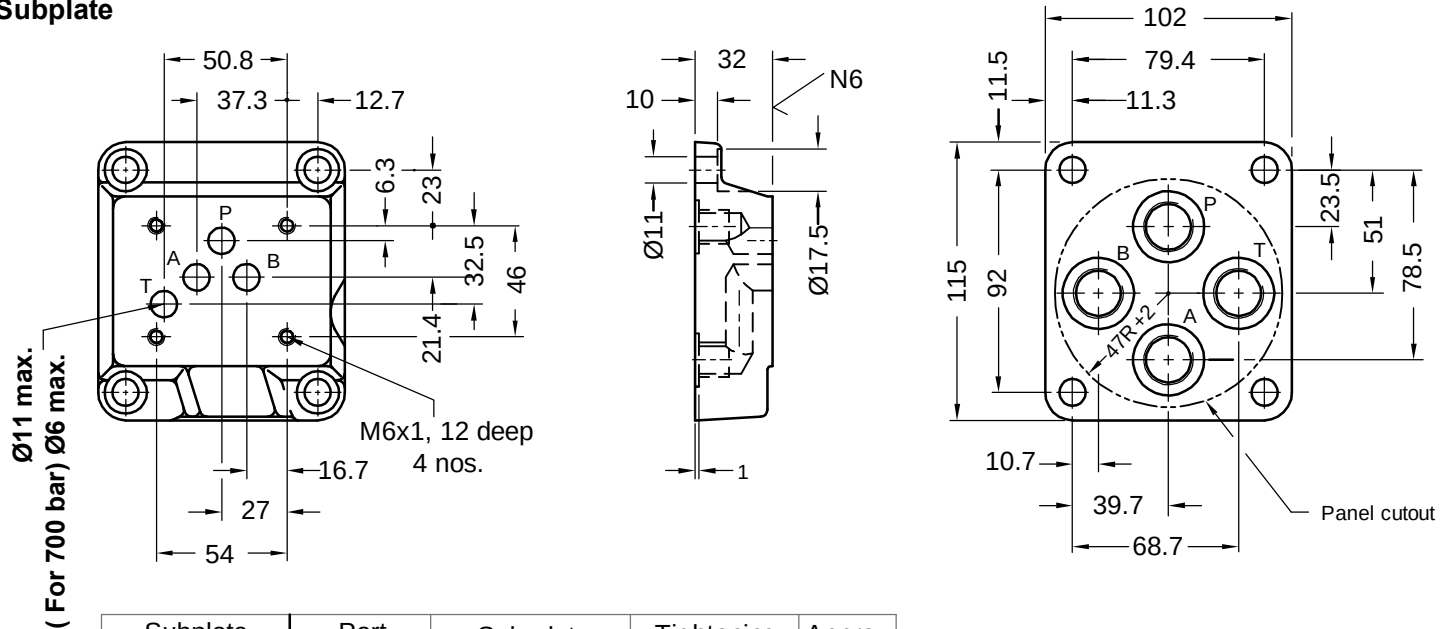
Subplate mounting body

Unit dimensions

Dimensions in mm.



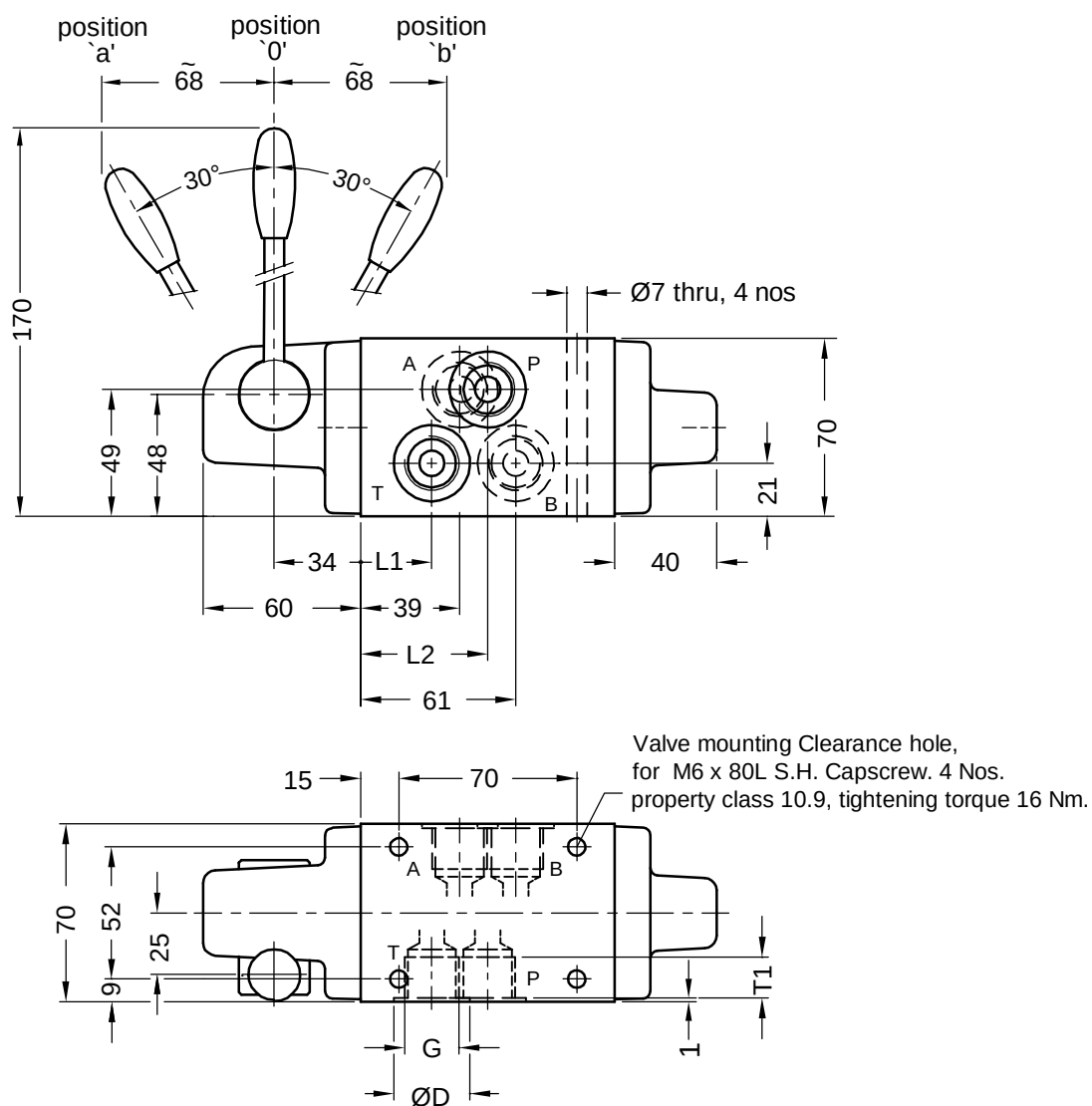
Subplate



Subplate Ordering Code	Port A, B, P, T	Sub-plate Fixing Screws	Tightening Torque	Approx Mass
G101	G 1/4"	M10 x 25 L	70 N-m	2.5 kg
G102	G 3/8"			
G103	G 1/2"			

Note
Sub-plate to be ordered separately.

Threaded mounting body



Ordering Code	Port Size G	ØD	T1	L1	L2
T02	G ¼"	22	12	28	50
T04	G ½"	30	16	31	54



Performance Curves for 4DL 10

(Pressure drop related to flow, with oil viscosity 46 cSt at 40°C)

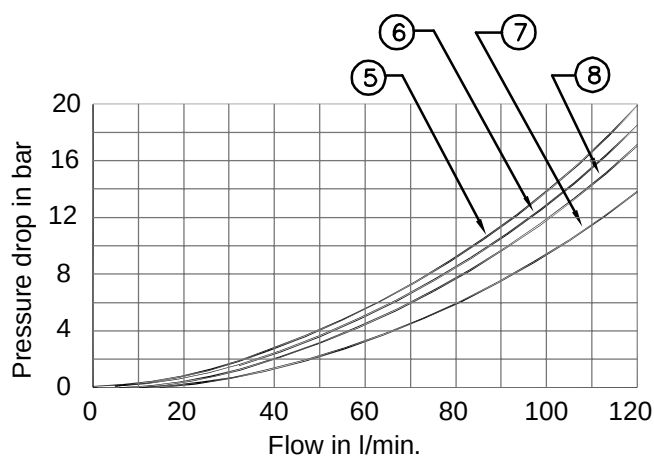
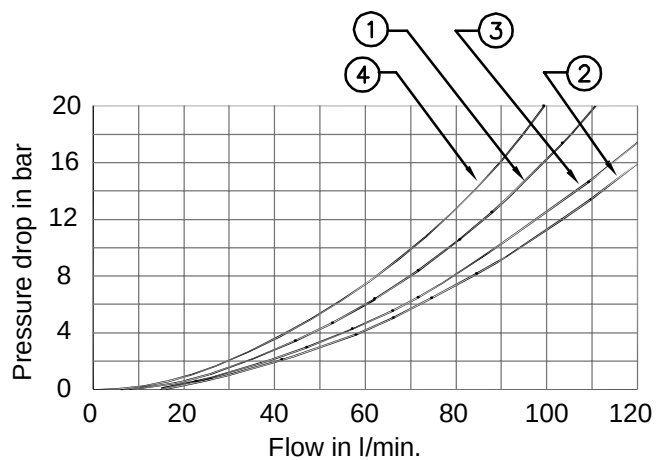
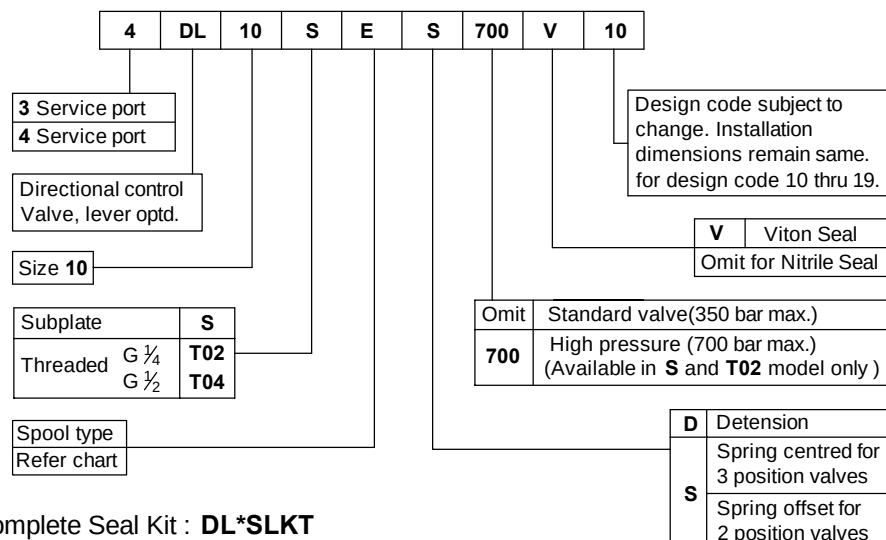


Table showing the relation between the spool type, direction of flow and curve graph to be referred to.

Spool type	Direction of flow / Curve no.				
	P to T	P to A	P to B	A to T	B to T
A	-	1	1	-	-
C	-	1	1	2	3
D	-	1	1	2	3
E	-	1	1	2	3
F	4	1	1	7	3
G	4	1	1	2	3
H	5	6	6	7	8
J	-	1	1	7	8
L	-	1	1	7	3
M	-	6	6	2	3
P	4	1	1	2	8
Q	-	1	1	2	3
U	-	1	1	7	8
W	-	1	1	2	3



Ordering code



Note

Ordering code for a complete Seal Kit : DL*SLKT

Spool Chart

Type	Symbol	Crossover	Type	Symbol	Crossover
	a 0 b A B P T	A B P T		a 0 b A B P T	A B P T
A			Q		
C			U		
D			W		
E					
F					
G					
H					
J					
L					
M					
P					



A Polyhydon Group Company

ROTARY DIRECTIONAL CONTROL VALVE Model : 4RDL

Ref. No. D 06429

Release 01 / 2026

ENGINEERING - 1 of 4

Description

Rotary directional control valve are hand lever operated valves designed for applications where,

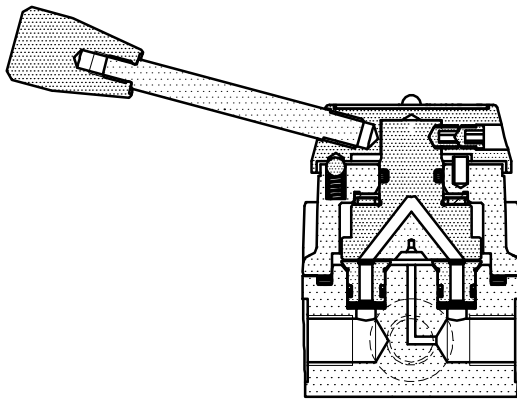
1. Pressures are high
2. Internal leakages in the valve need to be negligible and,
3. Flow handled are very small.

These valves are available with open or closed crossover condition while in operation.

Three mounting styles are designed to suit specific customer needs.



Section



Hydraulic symbol

Spool type	Basic symbol	Closed crossover (C)	Open crossover (O)
C		CC 	CO
E		EC 	EO
G		GC 	GO

Technical specifications

Construction	:	Rotary disc type.
Mounting	:	Threaded and subplate body.
Mounting position	:	Optional.
Interface	:	Factory standard.
Flow direction	:	Refer spool type given above.
Flow handling capacity	:	Refer graph shown.
Operating pressures	:	Ports

Open crossover
operating pressure (bar)

Closed crossover
operating pressures (bar)

P	700	500
T	10	10
A	700	500
B	700	500

Hydraulic medium	:	Mineral oil.
Temperature range	:	-20°C to + 80°C.
Viscosity range	:	10 cSt to 380 cSt.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Mass (approx)	:	Model : 4RDL02*T 4RDL02*S 4RDL02*SS
in Kg	:	1.5 1.6 2.1

Note : Torque required to operate the lever of closed crossover valve under pressure is approximately 14 Nm at 500 bar.

Torque required to operate the lever of open crossover valve under pressure is approximately 8 Nm at 700 bar.

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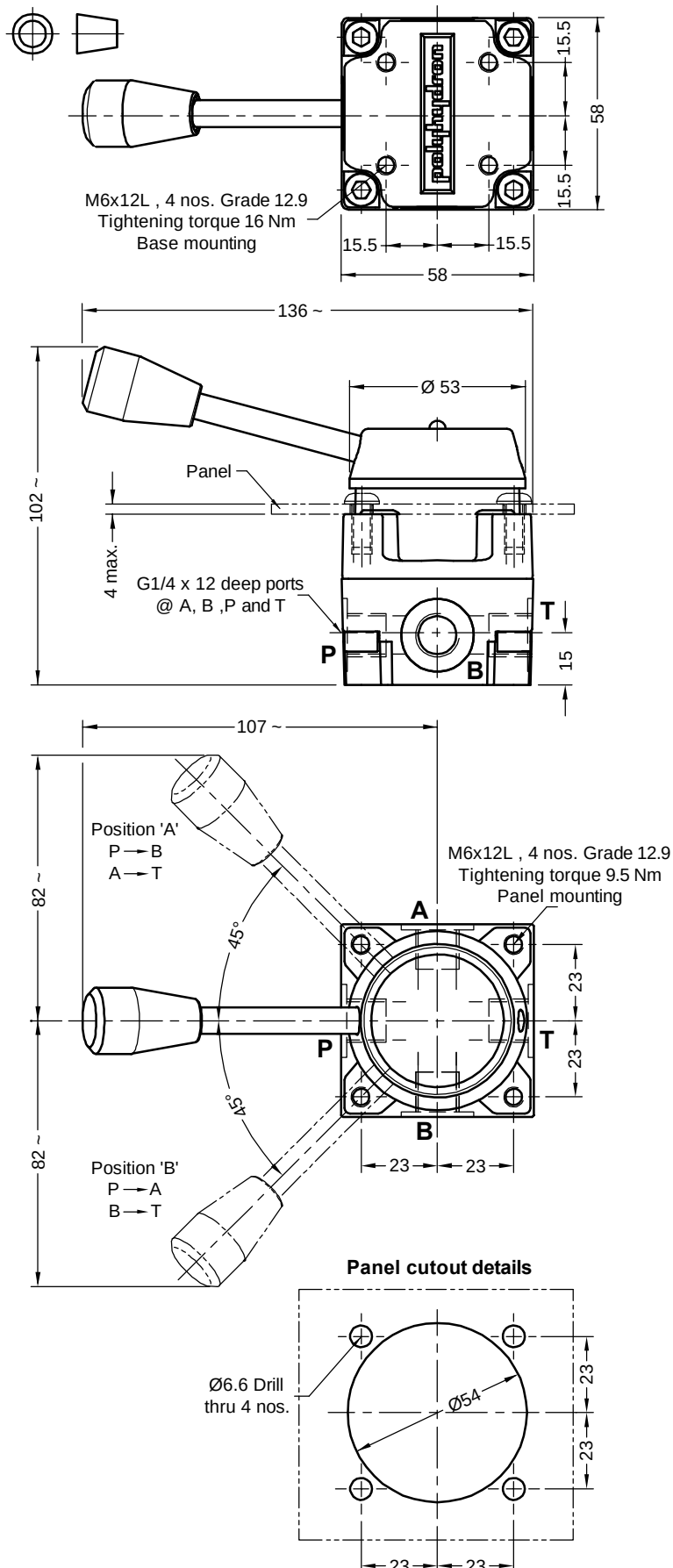
Phone : +91-(0)831- 2411001
E-mail : sales@polyhydon.com
Website : www.polyhydon.com



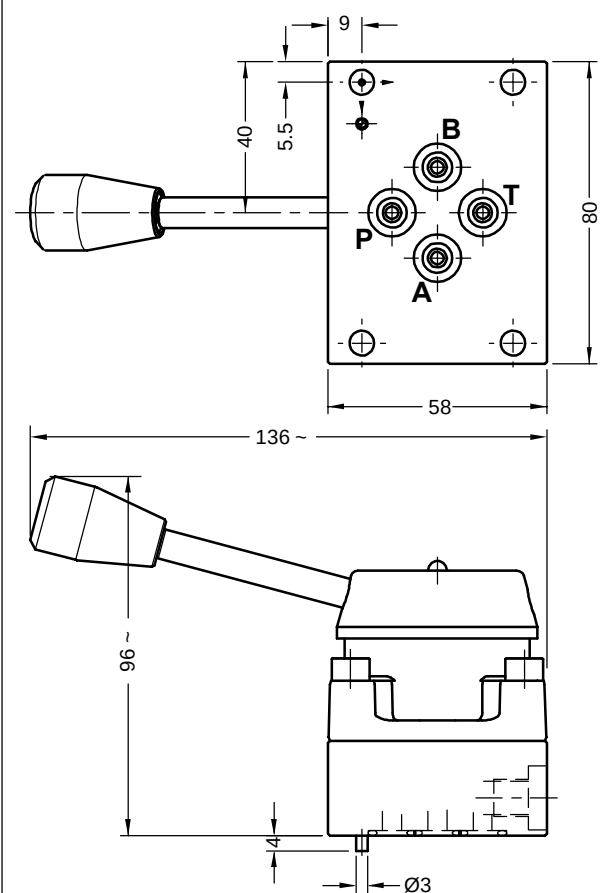
Unit dimensions

Dimensions in mm.

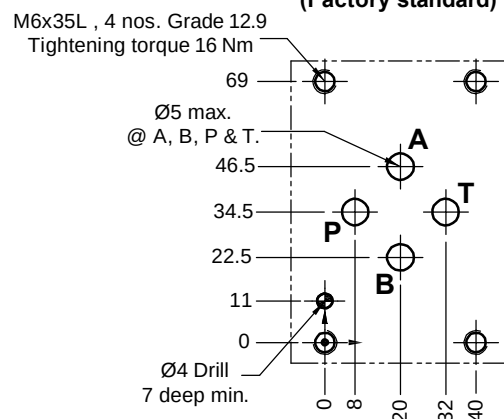
Threaded body



Subplate body

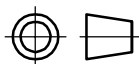


Subplate mounting interface (Factory standard)

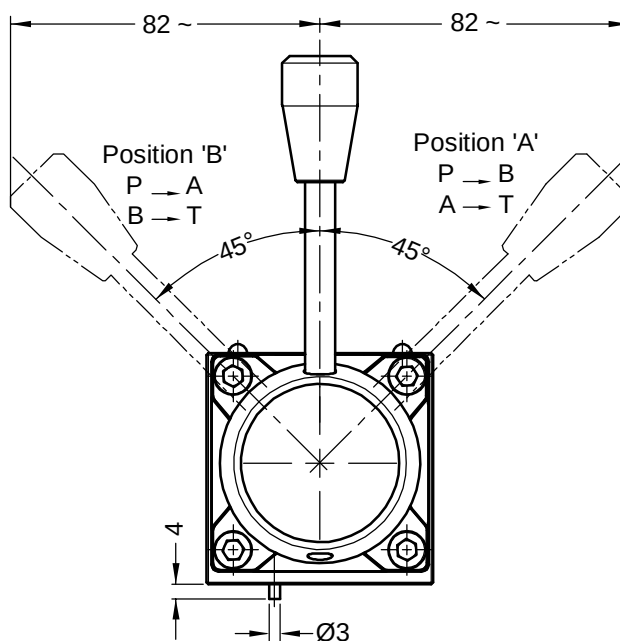
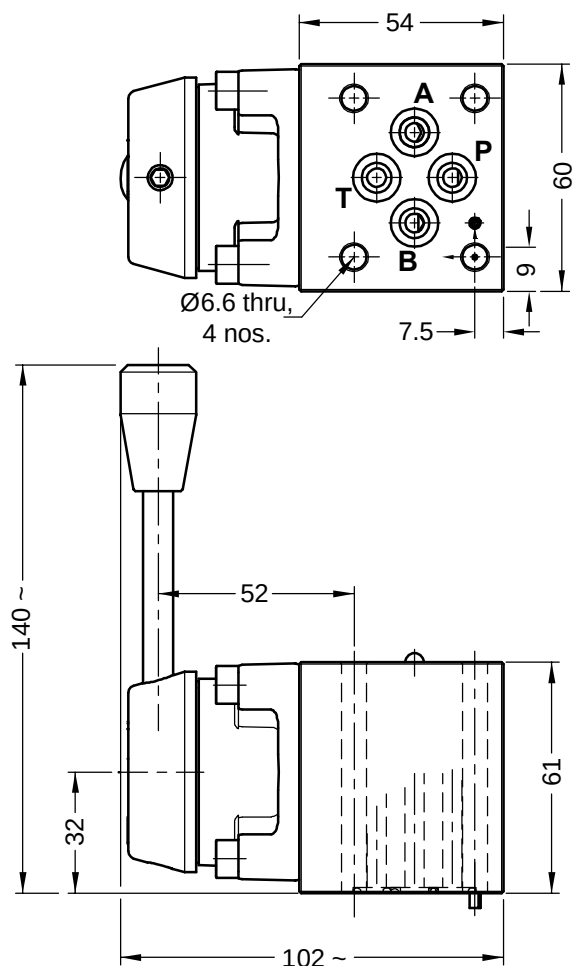


Unit dimensions

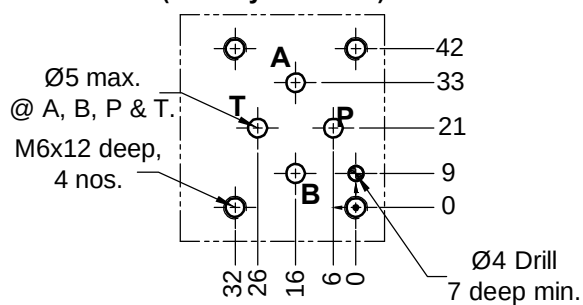
Dimensions in mm.



Side subplate mounting body



Side mounting subplate interface (Factory standard)

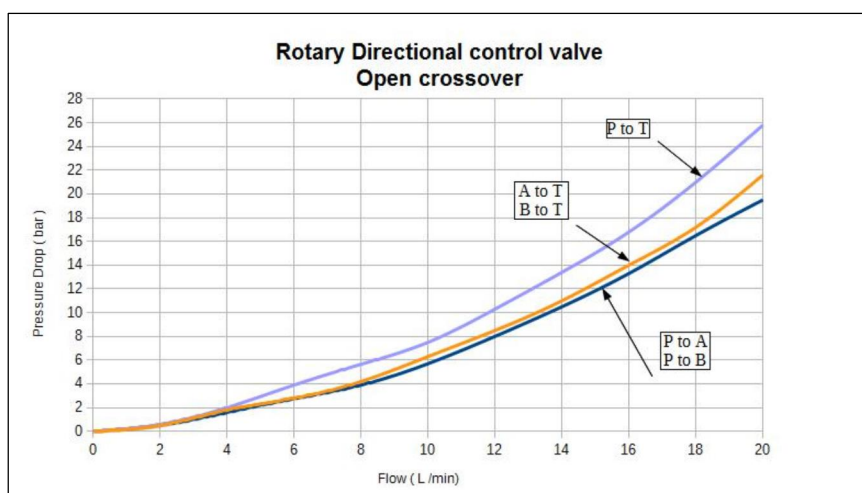
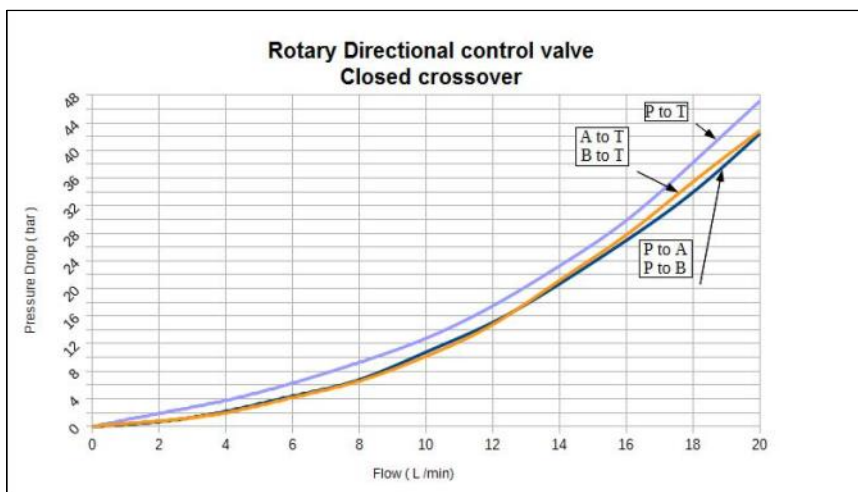


Valve mounting S.H.C Screws M6x70L (4 nos.)
Grade 12.9. Tightening torque 16 Nm.



Performance curve

Measured at temperature 40°C and with mineral oil of viscosity grade VG 46.



Ordering code

4	RDL	02	T	E	C	D	15
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4 Service ports

Rotary directional control valve lever operated

Size 02

Threaded

Subplate

Side subplate mounting

T

S

SS

Design code subjected to change. Installation dimensions remain same for design code 11 thru 19.

Detented

Spool type	Basic Symbol	C	O
		Closed crossover	Open crossover
C			
E			
G			

All valve fixing Screws are out of scope of supply.

Valve models 4RDL02HP**D-15 and 4RDL02HPT**D-15 are discontinued. For Rotary DCV with handpump refer datasheet no. D10180 and D 10100 in pumps section.